INCH-POUND

MIL-E-27669C(USAF)
15 September 1994
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3 November 1964

MILITARY SPECIFICATION

ENGINE INSTRUMENTATION SYSTEM, AIRCRAFT A/A24J-11

This specification is approved for use by the Department of the Air Force and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for one type of aircraft, engine instrumentation systems, designated A/A24J-11.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-1672	Boxes, Shipping, Reusable With Cushioning

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-V-173	Varnish, Moisture And Fungus Resistant, For Treatment Of Communications, Electronic And Associated Electrical Equipment
MIL-C-675	Coating Of Glass, Optical Elements (Anti-Reflection)
MIL-E-5400	Electronic Equipment, Aerospace, General Specification For
MIL-T-31000	Technical Data Packages, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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MIL-C-5541	Chemical Conversion Coatings On Aluminum And Aluminum Alloys
MIL-W-5846	Wire, Electrical, Chromel, And Alumel, Thermocouple
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification For
MIL-M-7793	Meter, Time Totalizing
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-S-20708	Synchros, General Specification For
MIL-G-26611	Generator, Tachometer GEU-7/A, Miniature
MIL-L-27160	Lighting Instrument, Integral, White, General Specification For
MIL-T-27179	Transmitter, Rate Of Flow TRU-27/A, Engine
MIL-B-27497	Bearing, Jewel, Sapphire Or Ruby, Synthetic
MIL-C-83488	Coating, Aluminum, Ion Vapor Deposited

STANDARDS

FEDERAL

FED-STD-101	Preservation, Packaging, And Packing Materials: Test Procedures
FED-STD-595	Colors

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-143	Specifications And Standards Order Of Precedence For The Selection Of
MIL-STD-471	Maintainability Demonstration
MIL-STD-701	List Of Standard Semiconductor Devices
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MIL-STD-2073-1	DOD Material Procedures For Development And Application Of Packaging Requirements
MS24264	Connectors, Receptacle, Electrical Flange Mount, Miniature
MS27034	Connector, Receptacle Electrical, Pin Insert, Cylindrical, Miniature Hermetic, Solder Mount
MS33558	Numeral And Letter, Aircraft Instrument Dial, Standard Form Of
MS91403	Cases, Large-Size (For Use With Electronic Equipment In Aircraft)
MS91405	Bases, Mounting, Large-Size (for Use With Electronic Equipment In Aircraft)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

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3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2

3.2 Components. The engine instrumentation system shall consist of the following components:

a. ECK-25/A24J-11 converter, fuel flow-tachometer-temperature data (one each)

b. EHK-19/A24J-11 indicator, temperature, synchro (two each)

c. ERK-9/A24J-11 indicator, electrical tachometer (four each)

d. EFK-18/A24J-11 indicator, rate of flow, fuel (two each).

3.2.1 Component compatibility and interchangeability. All system components specified in 3.2 shall be completely compatible and functionally interchangeable and shall be capable of meeting the demonstration tests specified in section 4 to determine compatibility and functional interchangeability of components manufactured by different suppliers.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials.

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent. If hermetically used within a hermetically sealed enclosure, fungicidal treatment will not be required. Where applicable, varnish in accordance with MIL-V-173 shall be used.

3.4.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to atmospheric and climatic conditions likely to be encountered in storage or normal service.

3..4.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.4 Magnesium alloy parts. Magnesium alloy parts shall not be used.

3.4.5 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts except where magnetic materials are essential.

3.4.6 Nonferrous materials. Nonferrous materials shall be used for all parts of the system except where ferrous materials are essential.

3.4.7 Toxic and corrosive fumes. The materials, as installed and under the service conditions specified, shall not liberate gases or fumes which will result in corrosion that will cause malfunction of equipment or discoloration of dials or indicia. Toxic gases or fumes that are detrimental to performance of the aircraft or health of personnel shall not be liberated under the service conditions specified herein.

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3.4.8 Protective treatment. When materials are used in the construction of the system that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4.9 Recycled and reclaimed materials. Recycled and reclaimed materials shall be encouraged to the maximum extent possible.

3.5 Design

3.5.1 Reliability. System reliability shall be a major design criterion. The system shall be so designed and constructed as to provide maximum practicable periods of time-between-failures. Failure is defined as any malfunction that results in degraded performance, regardless of the cause (careless handling or involuntary destruction excepted). The system shall have a minimum mean-time-between-failures (MTBF) of at least 652 hours based on the test specified in 4.6.3.13. The design goal of the system shall be a MTBF of 2,000 hours based on service experience.

3.5.2 Transistors and diodes. Transistors and diodes used in the system shall be selected in accordance with MIL-STD-701. Electron vacuum tubes shall not be used.

3.5.3 Power supply. The system shall be designed to operate from a 400-cycle aircraft alternator power supply rated 115V line to neutral in accordance with MIL-STD-704. Where any one instrument channel required power other than normal 115V 400-cycle, it shall be generated within the channel itself from the 115V 400-cycle power input to that channel. The synchro transmitters and control transformers required for the system shall conform to MIL-S-20708. Power failure in any one instrument channel shall not affect the power supply or normal operation of any other instrument channel related to that or any other engine.

3.5.4 Calibration adjustments. Calibration adjustments, where required, shall be located on the converter modules and shall be accessible without opening the module. A positive locking device shall be provided to prevent movement of the adjustment after it has been set.

3.5.5 Standard case. The ECK-25 converter shall be housed in a standard MS91403-B1D1 case and shall be equipped with an MS91405-B1D1 mounting base incorporating vibration isolators. Mounting-hole size and location shall be as shown on figure 1.

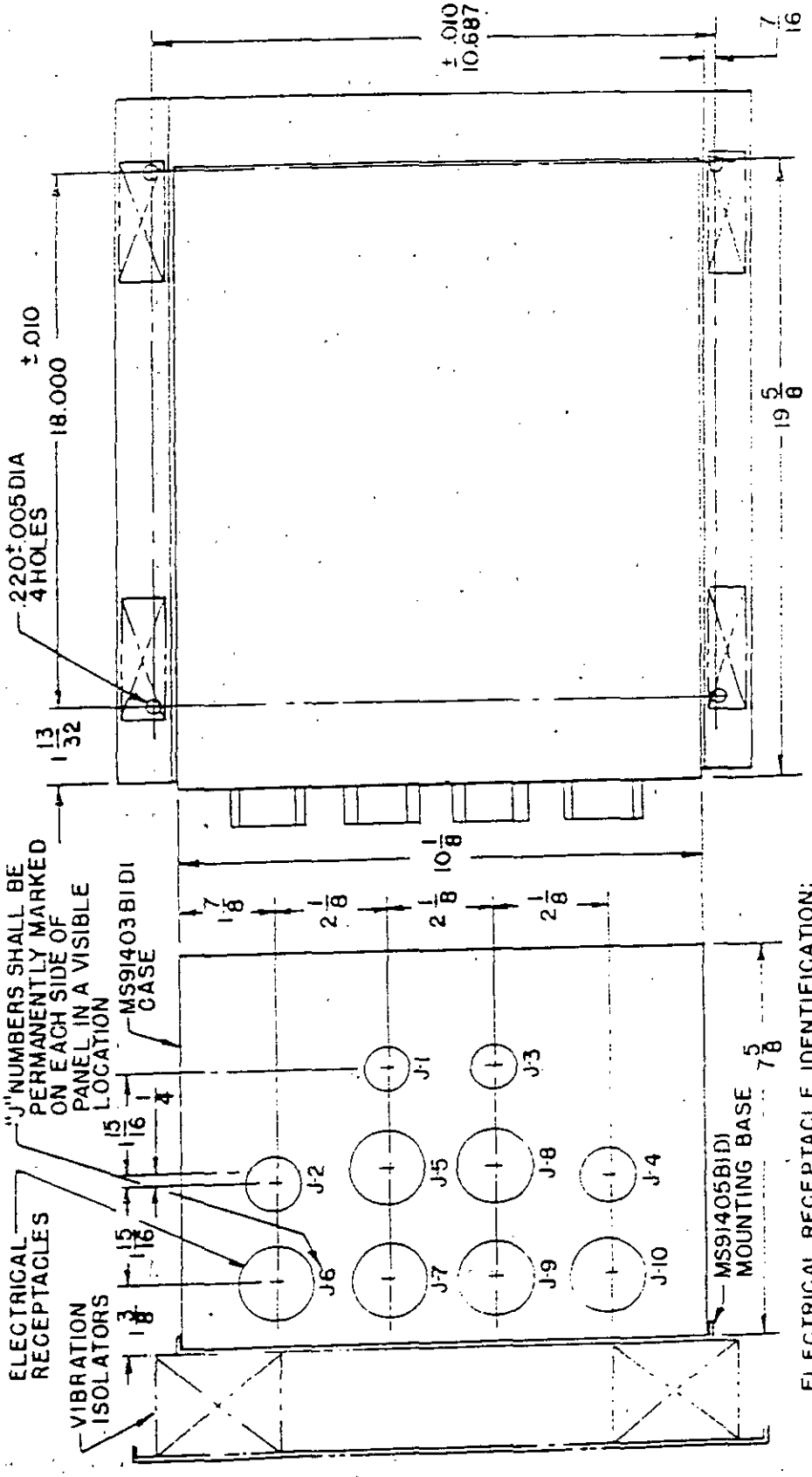
3.5.6 Module mounting. The design of the converter shall provide for a maximum of modular construction throughout the system for fast corrective maintenance by the substitution method. Each module shall be easily removed from the converter and replaced. Provisions shall be made for firmly fastening the modules in place. Each different type of module shall be keyed in a different way to prevent a module of a different type from being mounted in the wrong module bed. However, modules of the same type shall be mountable in any location suiting that type. The following components shall be put in separate removable modules and assemblies. Further modularization shall be subject to the approval of the procuring activity.

a. Each EGT converter channel (see 6.4.1)

b. Each N_1 and N_2 percent rpm channel (the N_1 channel is identical to the N_2 channel) and can be considered the same type as to part number and mounting

c. Each rate of flow module.

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ELECTRICAL RECEPTACLE IDENTIFICATION:

J-1	MS24264R14T15S
J-2	MS24264R18T31P
J-3	MS24264R16T24S
J-4	MS24264R18T14P
J-5	MS24264R22T55S
J-6	MS24264R22T55S6
J-7	MS24264R22T55S7
J-8	MS24264R22T55S8
J-9	MS24264R22T55S9
J-10	MS24264R22T55S10

FIGURE 1 CONVERTER OUTLINE

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3.5.7 Temperature ranges. The components of the system shall be designed to withstand the following ambient temperature ranges:

a. Converter -54° to +100° C operational to specified tolerances and -65° to +100° C exposure and operations

b. Indicators -54° to +71° C operational to specified tolerance and -65° to +71° C exposure and operation.

3.5.8 Malfunction influence. The design of the system shall be such that no single malfunction in one channel will affect any other channel in the system. A malfunction shall be considered any condition that would tend to cause any portion of the system to operate out of specified tolerance. Compliance with this requirement shall be demonstrated by a detailed fault analysis.

3.5.9 Dielectric strength.

3.5.9.1 Nonhermetically sealed components. Insulation for electrical circuitry and parts that carry 65V to 115V and are housed in nonhermetically sealed cases shall be sufficient to withstand a potential of 1,250V ac rms between the circuitry and the case. Insulation for electrical circuitry and parts that carry less than 65V shall be sufficient to withstand a potential of 500V ac rms between the circuitry and the case.

3.5.9.2 Hermetically sealed components. Insulation for electrical circuitry and parts that are housed within hermeticaaly sealed cases shall be sufficient to withstand a potential of 200V ac rms between the circuitry and the case. However, if the cases are unsealed for any reason, and the tests are performed with the cases unsealed, the electrical circuitry and parts shall meet the requirements for insulation within nonhermetically sealed cases.

3.5.10 Interconnecting wiring. Interconnecting wiring (converter to indicators, converter to input power, and converter to input engine parameters) shall be in accordance with figures 2, 3, 4, 5, 6, and 7.

3.5.11 Power consumption. The maximum power consumption for the converter and the eight indicators shall not exceed 300 va.

3.5.12 Maintainability. The following maintainability requirements shall be considered in the design of the system:

a. Minimization of the complexity of maintenance tasks (for example: calibration, adjustments, inspections, etc.) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items

b. Rapid and positive identification of the replaceable defective part, assembly, or component

c. Rapid and positive recognition of equipment malfunction or marginal performance

d. Minimum numbers and types of tools and test equipment (special and standard) needed to perform maintenance

e. Design to enhance and facilitate organizational and field level maintenance action

f. Require a minimum number and type of spare parts and assemblies.

3.6 Construction. The system shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipment, storage, installation, and service.

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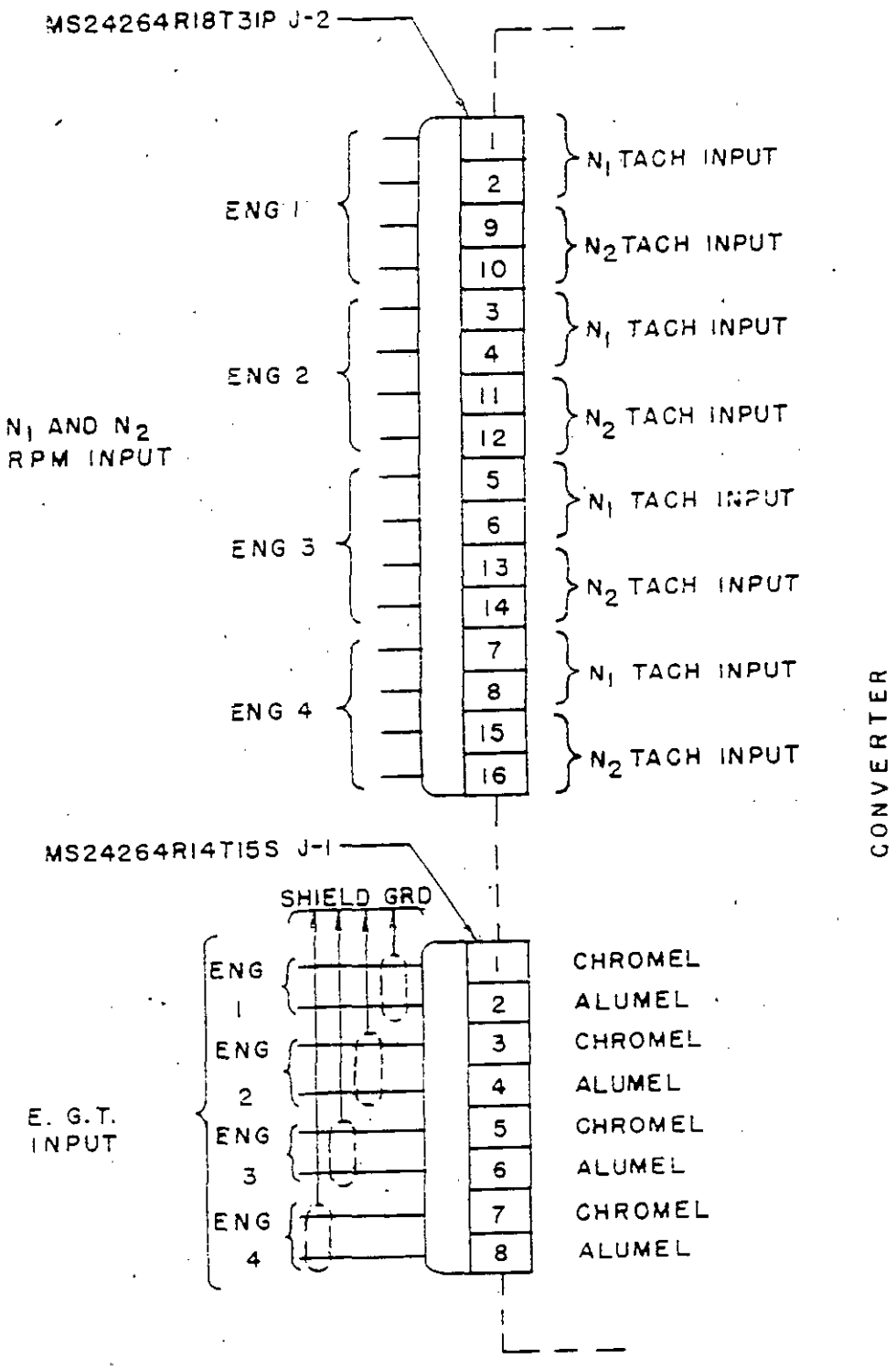


FIGURE 2. RPM and E.G.T input to converter wiring

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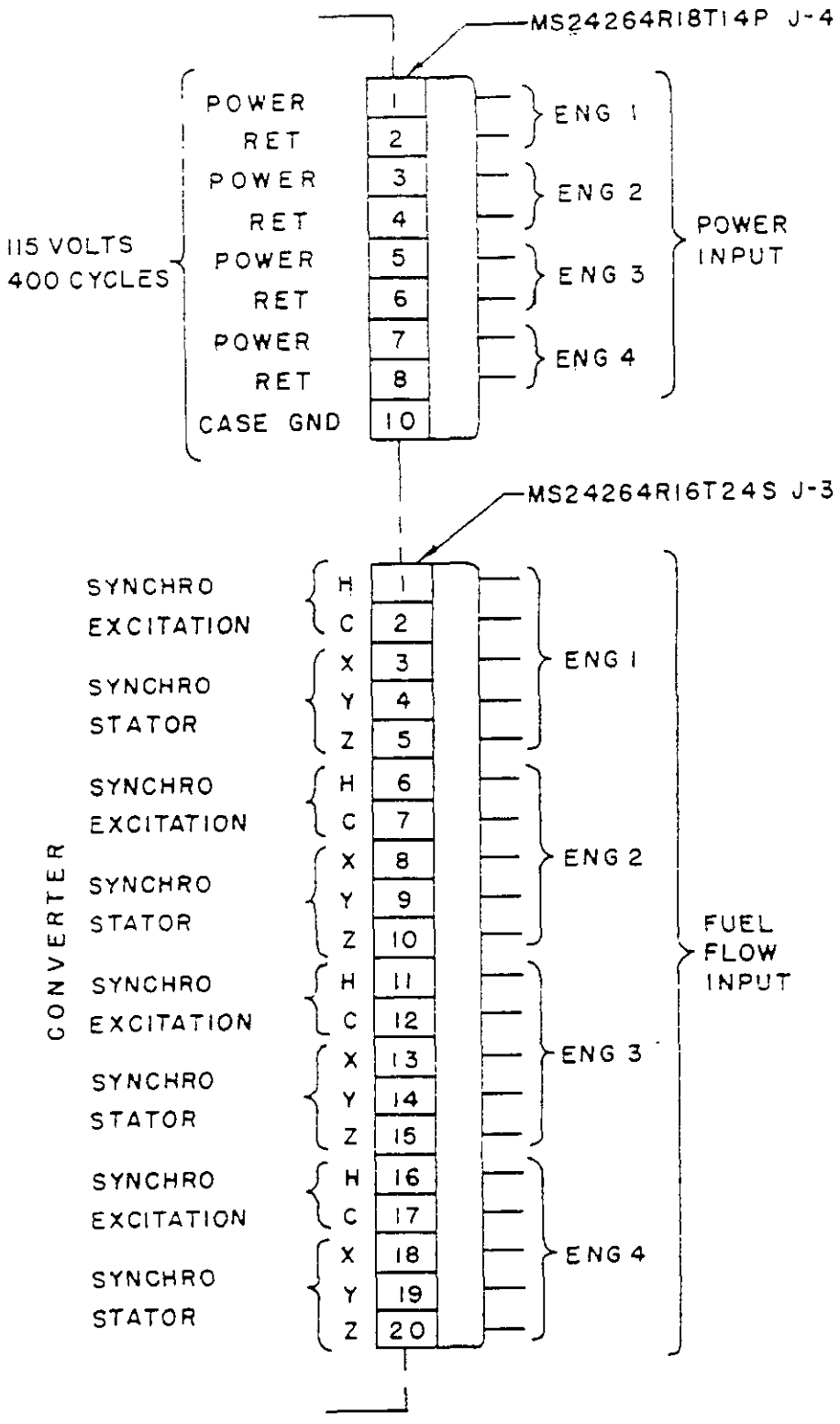


FIGURE 3. Power and fuel flow input to converter wiring

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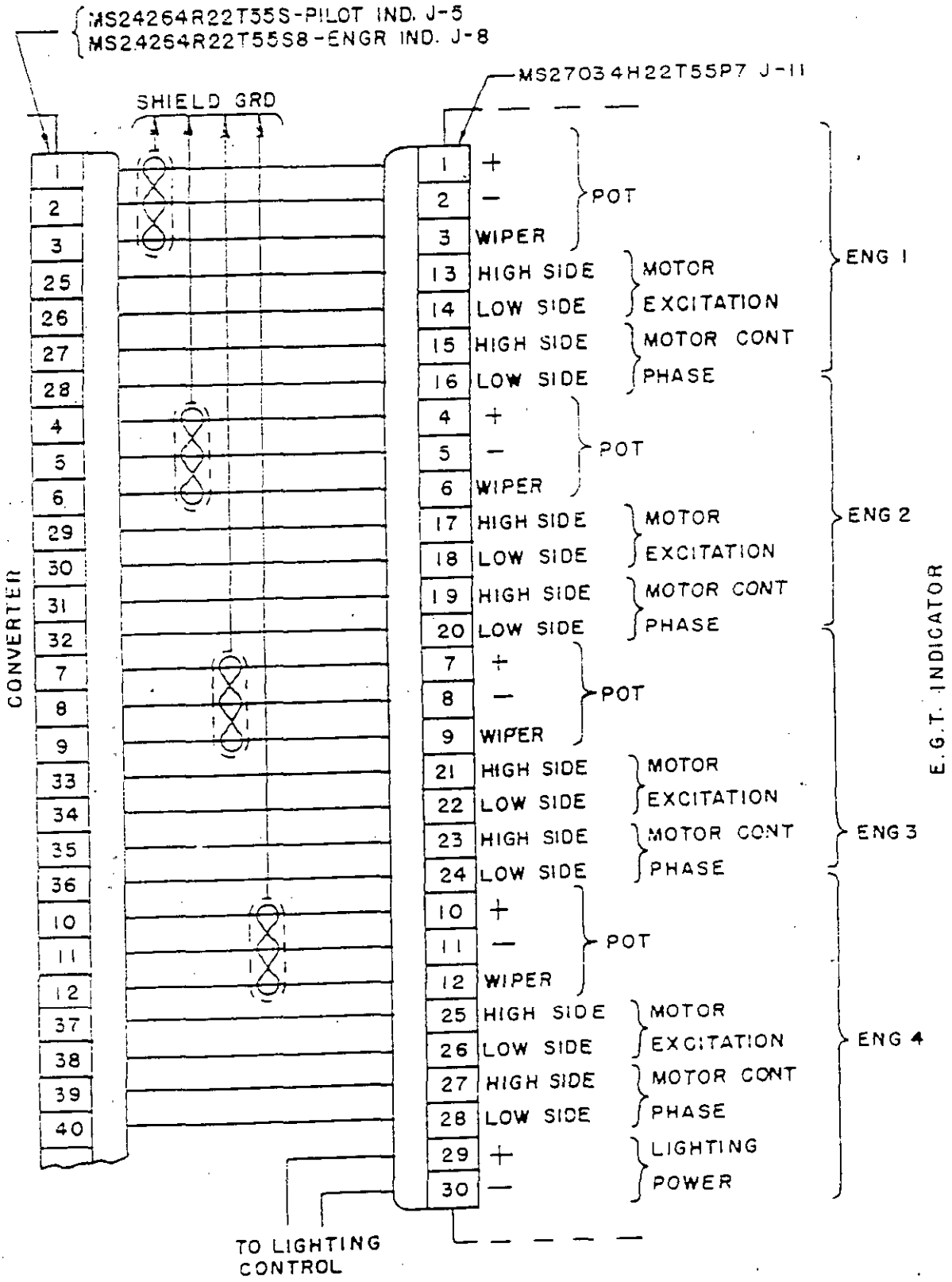


FIGURE 4. EGT, converter to indicator wiring

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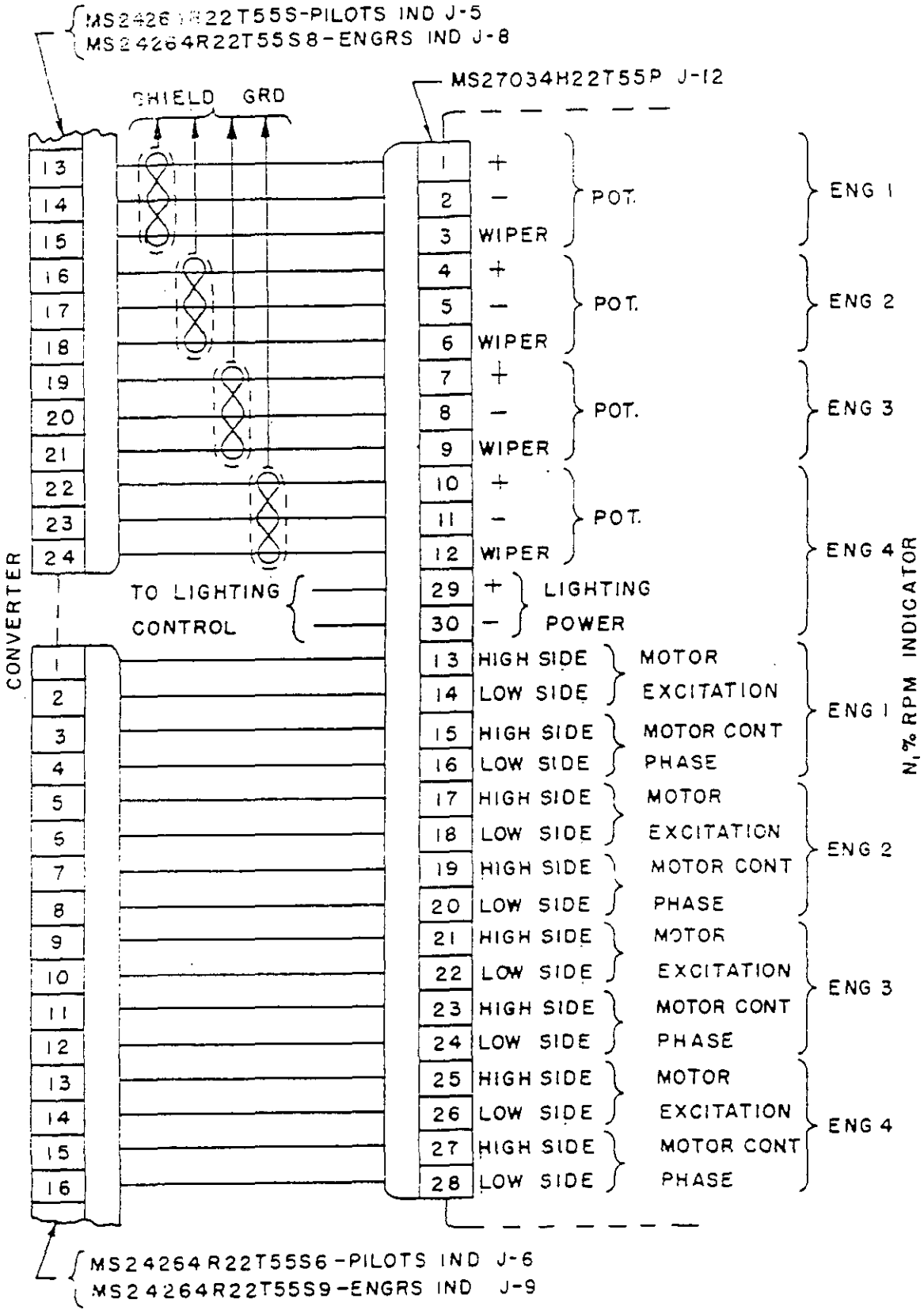


FIGURE 5. N, RPM, converter to indicator wiring

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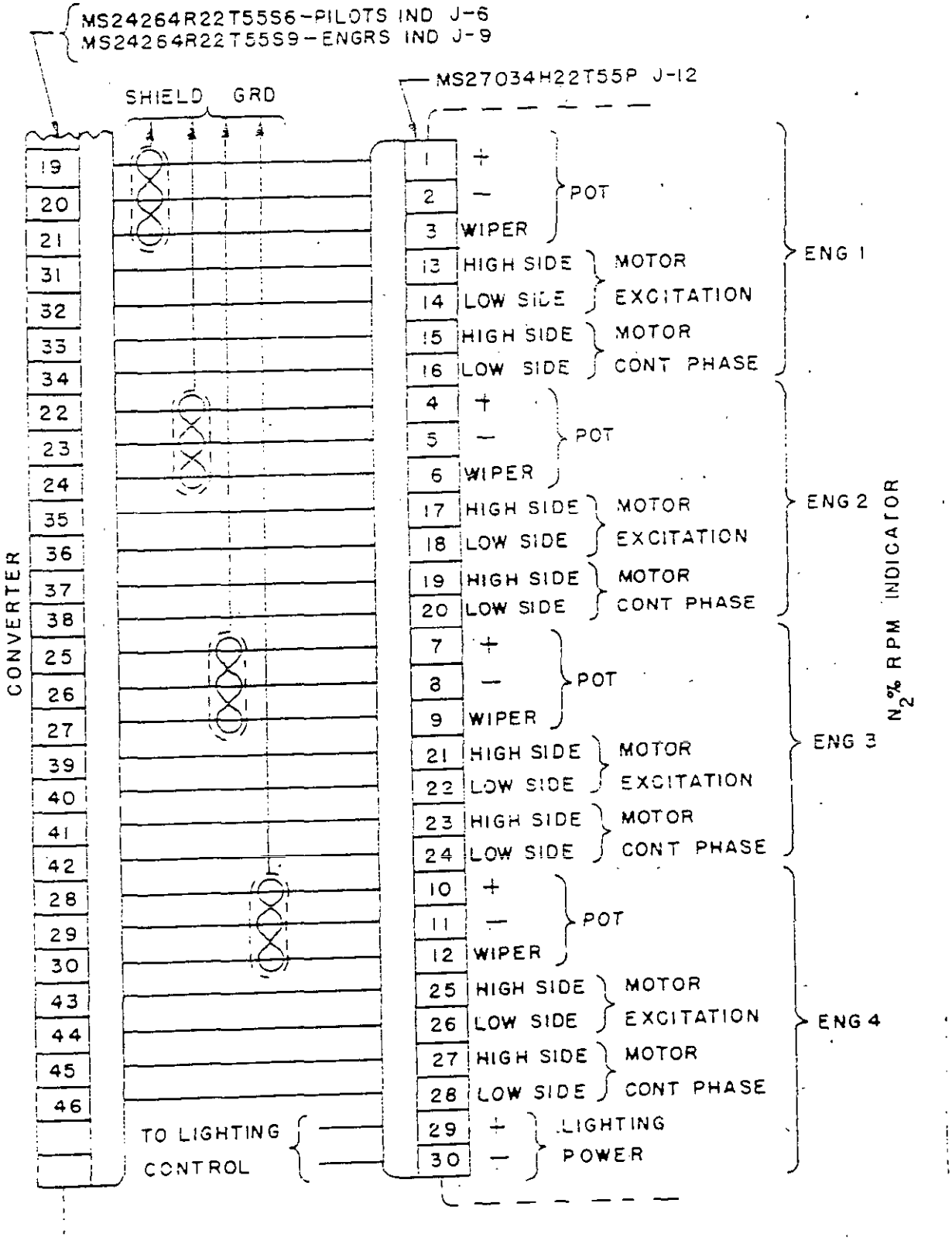


FIGURE 6. N_2 RPM converter to indicator wiring

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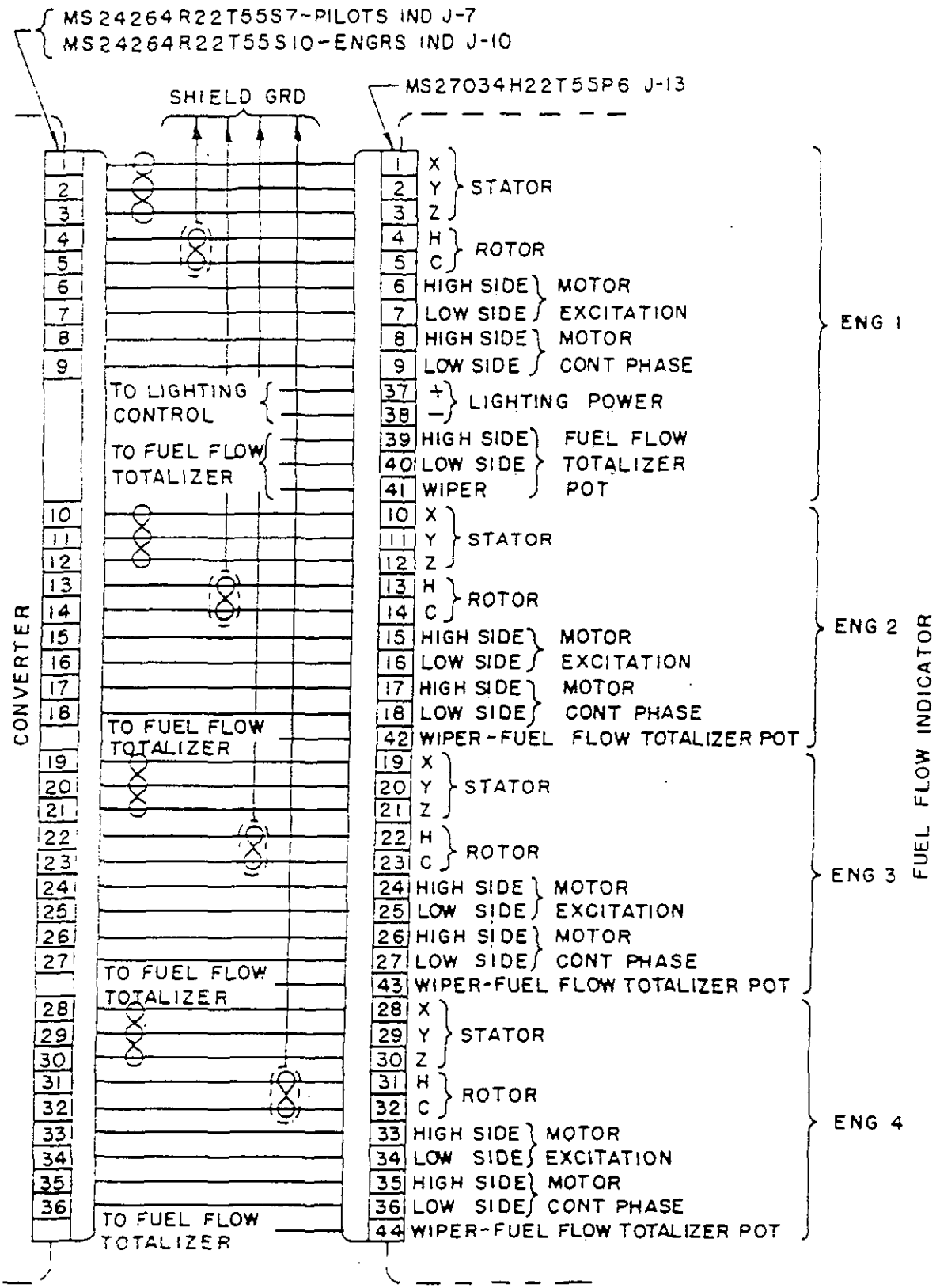


FIGURE 7. Fuel flow, converter to indicator wiring

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3.6.1 Pivots, bearings, and gears. Pivots, bearings, and gears shall not bind or shake, and shall be as frictionless as practicable. Gears shall have anti friction backlash provisions where required for accuracy.

3.6.2 Screw threads. Screw threads shall be in accordance with MIL-S-7742

3.6.3 Synthetic jewel bearings. When synthetic sapphire or ruby bearings are used, they shall be in accordance with MIL-B-27497.

3.7 Performance. The system shall be capable of meeting the requirements specified herein under the following conditions:

a. Temperatures in accordance with 3.5.7

b. Ambient pressures equivalent to those attained at altitudes of -1,000 to +60,000 feet

c. Salt fog in accordance with MIL-STD-810, Method 509.1, except as modified herein.

3.7.1 Sealing. For hermetically sealed units, leakage shall not exceed 1.0 or 10.0 micron cubic foot per hour at a pressure differential of approximately one atmosphere for the nominal 10 and 100 percent helium filling mediums, respectively.

3.7.2 System error. When tested as a system, the scale error shall not exceed the applicable combined tolerances listed in Tables I and II for the respective test conditions.

3.7.3 Component scale errors. When tested individually, the scale error tolerances of the converter shall not exceed those listed in Table I and the scale error tolerances of the indicators shall not exceed those listed in Table II.

3.7.4 Sensitivity. Sensitivity of the system shall be such that when input signal changes of 2° C. EGT, 0.25 percent rpm, and 10 pph fuel flow are applied to the respective converter inputs. The respective indicator readings will change as follows:

a. $2^{o} = 0.5^{o}$ C EGT

b. 0.25 = 0.12 percent rpm

c. 10 ± 5 pph fuel flow.

3.7.5 Warm-up time. The system shall perform in accordance with the requirements specified herein within one minute after power has been applied when the ambient temperature is above -20° C prior to excitation. For the temperature range of -54° to -20° C, the warm-up time shall not exceed two minutes.

3.7.6 Endurance. The system shall withstand at least 1,500 hours of electrical operation while undergoing 50,000 cycles of operation driving the indicators from approximately the low scale positions to 80 percent of the maximum positions and return at a rate of not less than 2 cpm.

3.7.7 Radio noise suppression. The system radio noise suppression requirements shall be in accordance with MIL-I-6181 for the frequency range of 0.15 to 1,000 mc. Where filtering is required, the volume and weight of filtering components shall be minimized by proper electrical and mechanical design and shall be contained within the housing. The system shall not be susceptible to electrical interference exceeding the limits defined in MIL-I-6181.

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TABLE I. Converter tolerances

Function and Test Point	Low and High Temperature	Room Temperature	Tape Oscillation Span	Tape Variation Span
(°C EGT)	(±°C EGT)	(±°C EGT)	(°C EGT)	(°C EGT)
0	6	5	2	2
100E	6	5	2	2
200	6	5	2	2
300E	4	3	2	2
400E	4	3	2	2
500	4	3	2	2
600E	4	3	2	2
700	4	3	2	2
(% RPM)	(±% RPM)	(±% RPM)	(% RPM)	(% RPM)
10	0.6	0.5	0.25	0.25
20E	0.6	0.5	0.25	0.25
30	0.6	0.5	0.25	0.25
40	0.6	0.5	0.25	0.25
50E	0.6	0.5	0.25	0.25
60	0.6	0.5	0.25	0.25
70	0.6	0.5	0.25	0.25
80	0.6	0.5	0.25	0.25
90E	0.4	0.3	0.15	0.15
100E	0.4	0.3	0.15	0.15
110	0.4	0.3	0.5	0.15

NOTES: Fuel flow test points and tolerances for the converter output shall be the same as for the rate of flow indicator specified in Table II.

E = Short scale error test points

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TABLE II. EGT indicator accuracy tolerances

TEST POINTS (°C)	SCALE SPECIFIED -55°, + 71° (°C)	ERROR (± DEVIATION TRANSMITTER INPUT) Room Temperature (°C)	FROM Position (°C)	Friction (°C)
00	7	5	1	1
100E	7	5	1	1
200	7	5	1	1
300E	4	2.4	1	1
400E	4	2.4	1	1
500P	4	2.4	1	1
600E	4	2.4	1	1
700	5	3	1	1

ELECTRICAL TACHOMETER INDICATOR ACCURACY TOLERANCES

TEST POINTS (% RPM)	SCALE SPECIFIED -55°, + 71° (% RPM)	ERROR (± DEVIATION TRANSMITTER INPUT) Room Temperature (% RPM)	Position (% RPM)	Friction (% RPM)
10	0.6	0.4	0.25	0.25
20E	0.6	0.4	0.25	0.25
30	0.6	0.4	0.25	0.25
40	0.6	0.4	0.25	0.25
50E	0.6	0.4	0.25	0.25
60	0.6	0.4	0.25	0.25
70	0.6	0.4	0.25	0.25
80	0.6	0.4	0.25	0.25
90E	0.3	0.2	0.25	0.25
100EP	0.3	0.2	0.25	0.25
110	0.3	0.2	0.25	0.25

RATE OF FLOW INDICATOR ACCURACY TOLERANCES

TEST POINTS (% RPM)	SCALE ERROR Degrees (Synchro Angle) Transmitter Setting	(± DEVIATION FROM TRANSMITTER INPUT -55°, + 71° (°C)	SPECIFIED IN PPH) Room Temperature (°C)	Position	Friction
400E	76.8	35	25	15	15
1,000	72.0	35	25	15	15
2,000	64.0	35	25	15	15
3,000E	56.0	35	25	15	15
4,000	48.0	35	25	15	15
5,000P	40.0	35	25	15	15
6,000E	32.0	35	25	15	15
8,000E	16.0	125	100	60	60
12,000E	344.0	125	100	60	60
16,000	312.0	125	100	60	60

NOTE: E = Short scale error test points.
P = Short friction error test points.

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3.7.8 External pressure. The converter and indicators shall withstand an external pressure of 56 inches Hg absolute without failure of the hermetic seals and indicator cover glass and without distortion of the cased that would affect normal operation.

3.7.9 Converter. The converter shall be capable of meeting the requirements specified herein when subjected to the following conditions:

a. Fungus growth in accordance with MIL-STD-810, Method 508.1

b. Dust (fine sand) in accordance with MIL-STD-810, Method 510.1, procedure I

c. Shock in accordance with MIL-STD-810, Method 516.2, procedure I

d. Humidity in accordance with MIL-STD-810, Method 507.1

e. Vibration in accordance with MIL-STD-810, Procedure I, curve D.

3.7.10 Converter response. The EGT and tachometer indicator converter channels shall not require more than five seconds to drive their applicable indicator form minimum to full scale when tested as specified herein.

3.7. 11 Indicators. The indicators shall be capable of meeting the requirements specified herein when subjected to the following conditions:

a. Vibration in accordance with MIL-STD-810, Method 514.2, procedure I, part 2 only, curve B

b. The indicators, both operating and nonoperating, shall withstand magnetic effect when positioned 5 $\pm$0.5 inches from a magnet that produces a field intensity of 0.17 to 0.19 oersteds at the indicator, without producing a compass deflection of more than 5°

c. Thermal shock in accordance with 4.6.2.10.

3.7.12 Indicator response. The indicators shall be capable of moving the tapes from minimum to maximum scale position in six seconds at room temperature and ten seconds at -54° and $+71^{o}$ C.

3.8 Converter. The converter shall provide for the following:

a. Four converter channels as specified herein, each of which shall drive two EGT indicator channels (see 6.4.3)

b. Eight converter channels as specified herein, each of which shall drive two tachometer indicator channels

c. Four converter channels, each of which shall receive synchro signals from a fuel flow transmitter and drive two rate of flow indicator channels as specified herein.

d. Power input (115V 400-cycle) to the EGT, rate of flow, and tachometer indicator channels.

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TABLE III. Channel parameters

PARAMETER	CHANNELS (PER ENGINE)	PROVISIONS FOR NO. OF ENGINES	TOTAL CHANNELS	INDICATORS DRIVEN BY ONE CHANNEL
Exhaust gas temperature	1	4	4	2
Percent RPM (N_1)	1	4	4	2
Percent RPM (N_2)	1	4	4	2
Fuel flow	1	4	4	2

3.8.1 Function. The converter shall receive output signals from each EGT, N_1 tachometer, and N_2 tachometer, and rate of flow indicator sensor or transmitter. The converter shall convert the EGT, N_1 tachometer, and N_2 tachometer indicator signals so that they may be utilized by the indicators; however, since the rate of flow signal is of the synchro type, it need not be converted. Each converter channel shall be self-sufficient, except that it may receive its power requirements from the power supply specified herein. No one channel shall depend upon any other channel or mechanization of the converter for operation, and malfunction of any one channel shall not adversely affect the operation of any other channel. Each channel shall be designed to drive the number of respective indicators shown in Table III. The converter shall be so constructed that no channel will cause signal interference in other channels.

3.8.2 Converter channel design. Each converter channel shall be designed to drive two of its respective indicator channels through dual output receptacles wired in accordance with figures 4, 5, 6, and 7. Systems performance requirements specified herein shall be met when the converter channels are thus connected.

3.8.2.1 EGT converter channels. Each EGT converter channel shall be designed to operate in conjunction with a chromel-alumel thermocouple circuit having a circuit resistance up to 200 ohms and temperature-electromotive force characteristics as specified in Table IV.

3.8.2.1.1 Thermocouple cold junction compensation. Each EGT converter channel shall be so compensated that the output is independent of changes in the thermocouple cold junction ambient temperature within the tolerance specified herein. The cold junction compensation shall be accomplished in the EGT converter channel.

3.8.2.1.2 Standard voltage. The voltage required for circuit balancing shall not be provided through the use of standard or dry cells. The voltage shall originate from the normal power supply.

3.8.2.2. N_1 and N_2 tachometer converter channels. The converter shall provide for eight separate tachometer indicator (rpm) channels. The N_1 and N_2 inputs shall be taken from the two ungrounded leads of the three-phase, ac, GEU-7/A tachometer generators conforming to MIL-S-26611. The tachometer load shall be an isolated lead and shall not be referenced to power ground. The generator signals shall then be converted into signals which are suitable for driving the tachometer indicators specified herein. The tachometer indication shall be based upon 100 percent being equal to 4,200 rpm rotational speed on the tachometer generator shaft.

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TABLE IV. Chromel alumel electromotive force versus temperature characteristics (0° C Reference junction)

Temperature °C	Electromotive Force Millivolts	Temperature °C	Electromotive Force Millivolts	Temperature °C	Electromotive Force Millivolts
0	0.00	400	16.40	800	33.30
10	0.40	410	16.82	810	33.71
20	0.80	420	17.24	820	34.12
30	1.20	430	17.67	830	34.53
40	1.61	440	18.09	840	34.93
50	2.02	450	18.51	850	35.34
60	2.43	460	18.94	860	35.75
70	2.85	470	19.36	870	36.15
80	3.26	480	19.79	880	36.55
90	3.68	490	20.22	890	36.96
100	4.10	500	20.65	900	37.36
110	4.51	510	21.07	910	37.76
120	4.92	520	21.50	920	38.16
130	5.33	530	21.92	930	38.56
140	5.73	540	22.35	940	38.96
150	6.13	550	22.78	950	39.35
160	6.53	560	23.20	960	39.75
170	6.93	570	23.63	970	40.14
180	7.33	580	24.06	980	40.53
190	7.73	590	24.49	990	40.92
200	8.13	600	24.91	1,000	41.31
210	8.54	610	25.34		
220	8.94	620	25.76		
230	9.34	630	26.19		
240	9.75	640	26.61		
250	10.16	650	27.03		
260	10.57	660	27.45		
270	10.98	670	27.87		
280	11.39	680	28.29		
290	11.80	690	28.72		
300	12.21	700	29.14		
310	12.63	710	29.56		
320	13.04	720	29.97		
330	13.46	730	30.39		
340	13.88	740	30.81		
350	14.29	750	31.23		
360	14.71	760	31.65		
370	15.13	770	32.06		
380	15.55	780	32.48		
390	15.98	790	32.89		

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3.8.2.3 Rate of flow circuits. The converter shall accommodate four separate identical rate of flow input circuits. The rate of flow inputs shall be synchro signals provided from the TRU-27/A transmitter in accordance with MIL-T-27179. These inputs shall not be converted. Each input circuit shall be fed out as two identical single synchro outputs, each of which shall be capable of driving one rate of flow indicator. System accuracies specified herein shall hold when either one or both outputs are utilized. When the rate of flow synchro inputs provided from the TRU-27/A transmitter are signaling an increasing fuel flow value, the rate of flow indicators shall register compatible increasing readings. The same relationship shall apply for decreasing fuel flow signals. The converter shall supply instrumen- tation power to each circuit. Power requirements of the TRU-27/A transmitter shall be as specified in MIL-T-27179. The synchro zeroing procedure for the fuel flow indicator shall also be as specified in MIL-T-27179.

3.8.2.4 Converter channel outputs. Each converter channel output shall be such that it will drive one channel or a quadruple vertical scale, movable tape indicator of its respective type. The indicator mechanisms shall be servo driven and designed as specified herein. Due to space limitations in the indicator, the indicator shall contain only that part of . . .

the servo mechanism that is essential for driving the tapes. The converter module shall contain the bulk of the servo system, leaving only one servo motor and its corresponding follow-up device in each channel of the EGT and tachometer indicators, and the synchro-controlled, servo-driven indicator channels for the rate of flow indicators.

3.8.2.5 Power-off circuit. The converter shall be equipped with power-off circuits which shall actuate the power-off warnings in the indicators. Each indicator channel shall have its own power-off detection circuit which shall operate the power-off warning in its respective indicator channel only. The detection system shall show: "OFF" when any segment of power within the channel (except for lighting) is absent either because of failure of the *special power generated within the channel or due to lack of main input power.*

3.8.2.5.1 Circuit protection. Sufficient circuit protection shall be provided as part of the converter so that short circuiting of any one indicator parameter circuitry associated with any one engine will not deprive any other *indicator circuitry of its power supply. The circuit protection devices shall be accessible from the outside of the* converter. The current ratings of fuses, if used, shall be permanently marked on the converter adjacent to each fuse holder. The following shall be the minimum circuit protection required (see figure 8):

a. Separate protection for each *115V 400-cycle input to each EGT channel*

b. Separate protection for each 155V 400-cycle input to each N_1 rpm channel

c. Separate protection for each *115V 400-cycle input to each* N_2 rpm channel

d. Separate protection for each 115V 400-cycle input to each rate of flow channel.

3.8.2.6 Converter receptacles. *All connector inserts in the exhaust gas temperature thermocouple circuitry shall* be chromel and alumel material, as appropriate, to match the chromel and thermocouple input leads. The receptacles shall be located as shown on figure 1. The converter input and output receptacles shall be as follows:

a. EGT input - *MS24264R14T15S, except that four each chromel and alumel pins shall be incorporated -* (see figure 2)

b. N_1 and N_2 percent RPM input - MS24264R18T31P

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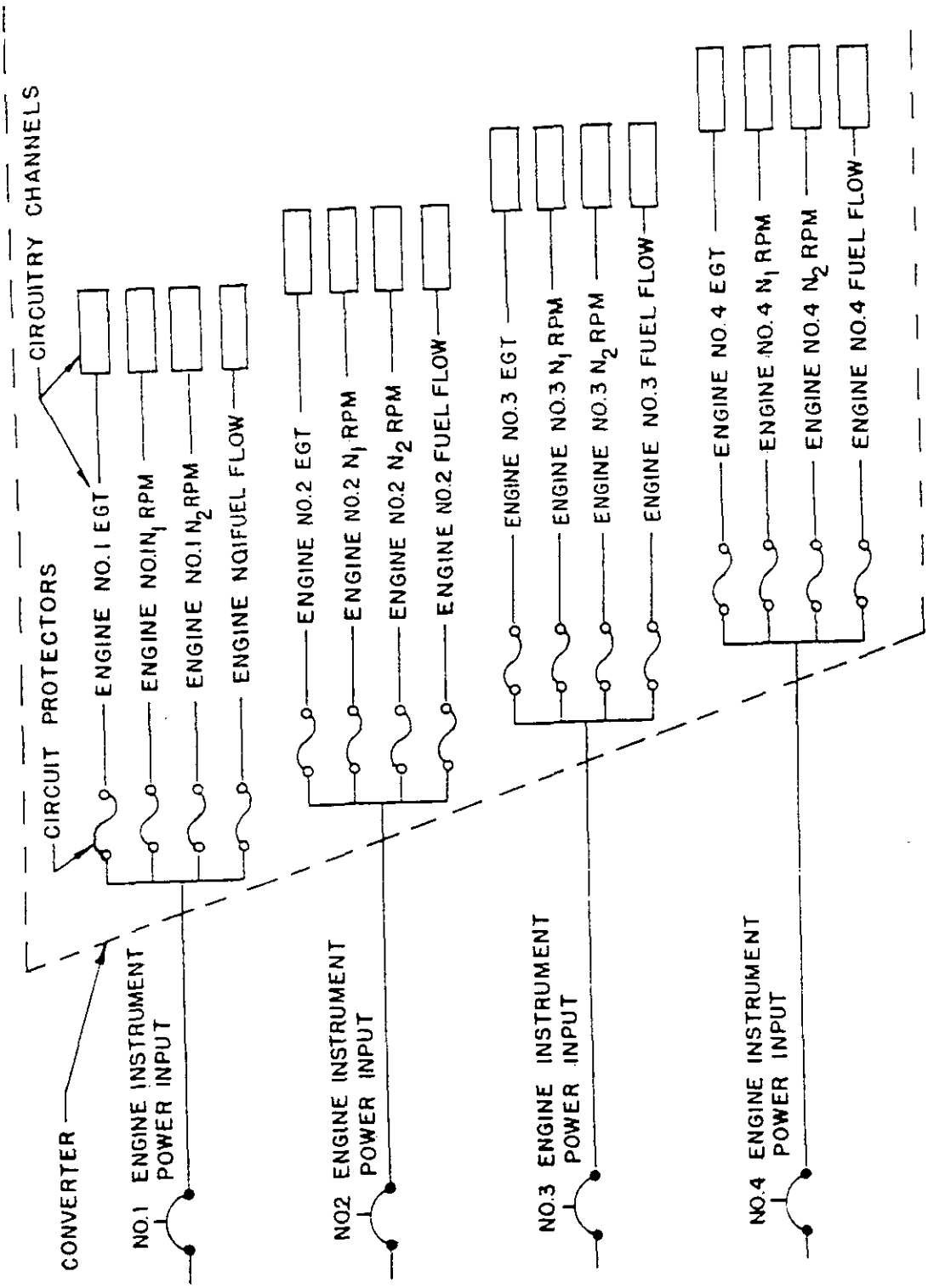


FIGURE 8 MAIN POWER SUPPLY CIRCUIT PROTECTION

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c. Rate of flow input - MS24264R16T24S

d. Electrical power input - MS24264R18T14P

e. Output circuitry of converter to be distributed among the following six receptacles:

(1) MS24264R22T55S

(2) MS24264R22T55S6

(3) MS24264R22T55S7

(4) MS24264R22T55S8

(5) MS24264R22T55S9

(6) MS24264R22T55S10.

3.8.2.7 Elapsed time indicator. A 10,000 hour , four-digit type elapsed time indicator conforming to MIL-M-7793 shall be provided and mounted in the converter so that the dial is visible on either the side or top of the case. The indicator shall function when power is applied to the converter.

3.9 Indicators. Each indicator shall provide for indication of four separate channels of the same parameter. Each indicator channel shall act independently and shall not depend upon the function of any other channel for operation. Not one channel shall adversely affect the operation of any other channel. Parameter indication shall be in the form of a servo-driven tape against a fixed scale as shown on figure 4. The channel mechanism shall align the top limit of the white portion of the tape with respect to the fixed graduated scale in accordance with the value dictated by the signal input to the respective converter channel and to the accuracies specified in Table II. The indicator shall be designed to employ shielded wire in its sensitive input circuits with the shields being at ground potential. Accuracies shall hold for all possible interconnecting lead lengths up to 100 feet (No. 22 wire, minimum size) separation between the indicator and its converter channels. The tape drive mechanism shall be equipped with stops or other suitable devices to prevent damage to the indicator by signals attempting to drive the tape beyond the upper and lower limits.

3.9.1 Lighting. The indicator shall be integrally lighted in accordance with MIL-L-27160. Lighting capability shall be based on visual comparison with a prime standard indicator.

3.9.1.1 Prime standard. The prime standard lighting reference instrument shall be prepared as follows:

a. A minimum of 20 areas covering the entire face of the indicator shall be measured for brightness level in foot-lamberts

b. The prime standard instrument and its recorded overall brightness measurements shall be approved by the procuring activity

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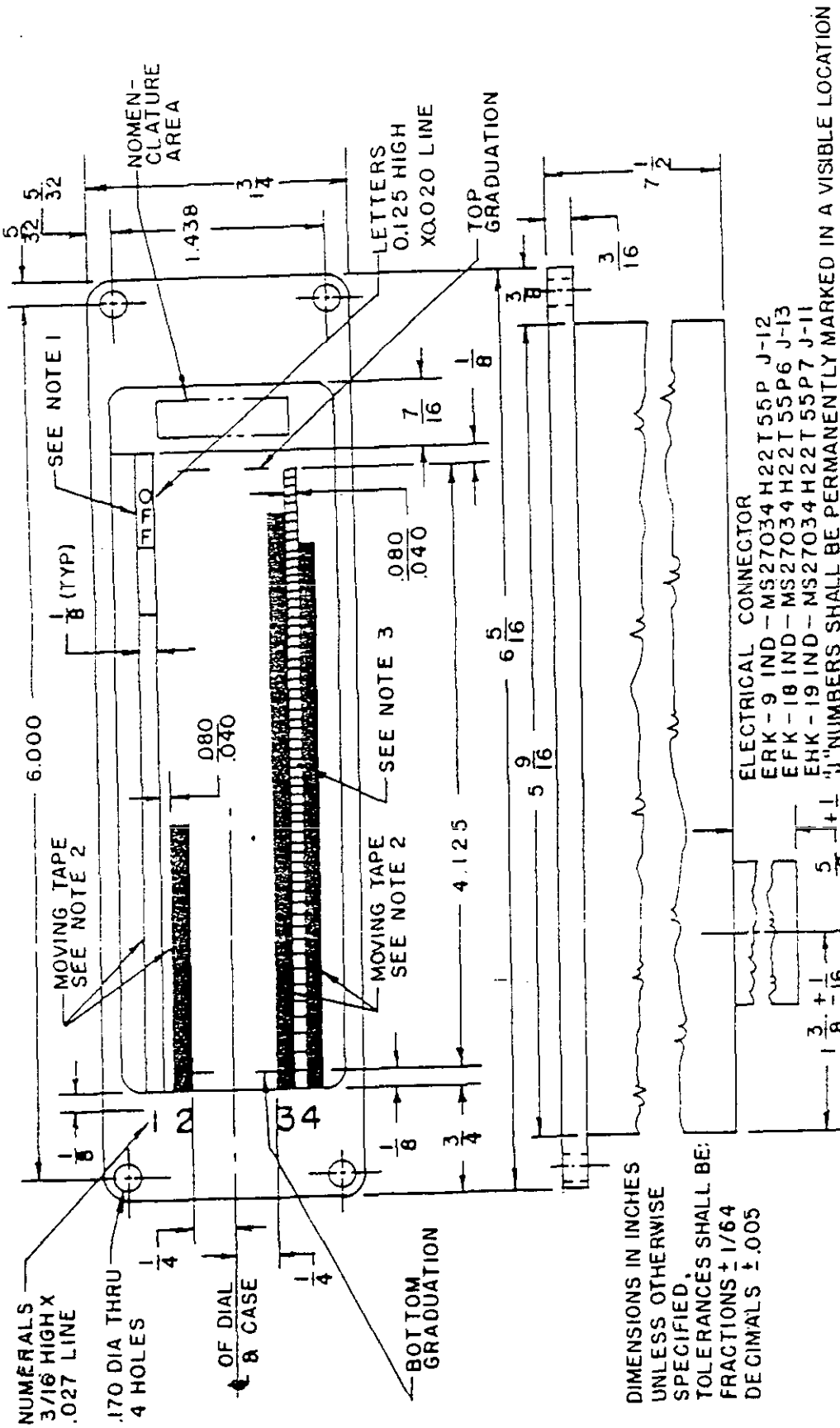


FIGURE 9 INDICATOR OUTLINE

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c. The standard shall be checked periodically, at least every 100 hours of operation, to determine any change in brightness

d. Variations greater than ± 20 percent in two or more areas will require return of the standard for corrective action

e. If a tape instrument is used as the prime standard, the tapes shall be fixed and shall be positioned in such a manner that a full scale reading and an "OFF" scale reading will be presented.

3.9.2 Indicator case. The indicator case design shall be in accordance with figure 9. It shall be constructed of a suitable material applicable to hermetic sealing (see 6.4.2) and shall be hermetically sealed. The material shall be uni- form in texture with a smooth external surface. The design of the case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed without the use of special tools and fixtures. Tear strips are provided for this purpose. Hermetic sealing shall be so accomplished that the seal will not be dependent upon materials that will be affected by the action of any atmosphere to which the indicator may be subjected.

3.9.2.1 Filling medium. The case shall be filled 100 percent nitrogen. This dry nitrogen filling gas must be at least 99.99 percent pure, free of dust particles, contain not more than 0.002 percent free oxygen and have a dew point not higher than -59.4° C (-75° F). The absolute pressure of the filling medium in the case shall be approximately one atmosphere.

3.9.3 Decompression. The indicators shall be so designed and constructed as to withstand explosive decompressions of the surrounding atmosphere. Operation shall not be affected when a pressure differential of 16 inches Hg is applied instantaneously between the interior and exterior of the case.

3.9.4 Indicator mechanisms. The indicator mechanisms shall be sufficiently rugged that small distortions of the cases will not affect the accuracy or operation of the mechanisms.

3.9.5 Dial. The respective dial layouts and markings shall be as shown on figure 10. The circuitry for the scale nonlinerities specified on figure 10 shall be provided for, either in the indicator or in its respective converter module.

3.9.6 Scales. The scales shall conform to figure 10. The scale face shall be securely attached to the frame of the indicator mechanism.

3.9.6.1 Scale markings. The scale graduations and dimensions of the markings shall be in accordance with figure 10. The markings shall be white, color No. 37875 of FED-STD-595. Dial numbers and letters shall to conform to MS33558.

3.9.6.2 Cover glass distance. The distance between the scale face and the inside of the cover glass shall be the minimum practicable and shall not exceed 0.100 inch.

3.9.6.3 Scale visibility. All scale markings and the total width of all tapes shall be visible from any position in front of the dial ranging between normal and a position 45° to the normal.

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3.9.7 Electrical connector. The indicators shall be provided with an electrical connector in accordance with figure 9 and MS27034. A means shall be provided for safety wring the interconnecting wiring connectors to the indicators and shall be subject to the approval of the procuring activity.

3.9.8 Power-off warning. Each tape in the indicators shall incorporate a power-off warning as shown on figure 9 which shall operate in conjunction with the power-off circuits specified in 3.8.2.5. Each servo shall be mechanized so that upon removal of the input power to a particular engine parameter, that tape will drive to an off-scale position in the direction of the decreasing function and display the word "OFF" within two seconds. The tape shall display the word "OFF" in black letters, Color No. 17038, FED-STD-595, on a red background. The background color shall be red fluorescent No. 116-1548 (Switzer Brothers, Cleveland, Ohio) or equal. The warning shall disappear from view within one second after input power is applied.

3.9.9 Tape drive stops. The electrical tachometer indicator drive mechanism shall incorporate lower and upper stops that will limit tape travel from "POWER-OFF WARNING" to a minimum of 0.031 inch above 110 percent rpm on the indicator dial. The rate of flow indicator drive mechanism shall incorporate lower and upper stops that will limit tape travel from "POWER-OFF WARNING" to a minimum of 0.031 inch above 16,000 pph on the indicator dial.
The exhaust gas temperature indicator drive mechanism shall incorporate lower and upper stops that will limit tape travel from "POWER-OFF WARNING" to a minimum of 0.031 inch above 700° C on the indicator dial.

3.9.10 Off-scale indication. Since tape travel is permitted below the lowest visible graduation, the top (black portion) of the rate of flow indicator tapes and the exhaust gas temperature, indicator tapes shall be equipped with a series of 0.047 inch diameter white dots to indicate movement of the tapes. The distance between the center of the dots shall be equal to the distance between the lowest increments of scale graduations at the bottom of the indicator scales. The position of the lowest dot on the tapes shall be such that it will be at the topmost scale graduation when the top of the white portion of the tapes is even with the lowest scale graduation (dots shall not be visible when the tapes are at normal operating positions). There shall be a sufficient number of dots on each tape to permit indication down to zero pph fuel flow and -60° C EGT, respectively, as they are read against the top scale graduation.

3.9.11 Fuel flow output. Each fuel flow indicator channel shall contain an output potentiometer. The wiper of each potentiometer shall be positioned to provide a voltage output directly proportional to the indicated rate of flow

for the respective channel when excited externally. The eight potentiometer excitation terminals shall terminate on two pins on each rate of flow indicator connector. The four wiper terminals shall be brought out to four pins on each rate of flow indicator connector. The output range of each potentiometer shall be from 0 - 16,000 pph. The potentiometer shall have the following characteristics:

a. Resistance 5K ohms ±5 percent

b. Power rating - 1.5w at 25° C

c. Linearity tolerance ±0.2 percent

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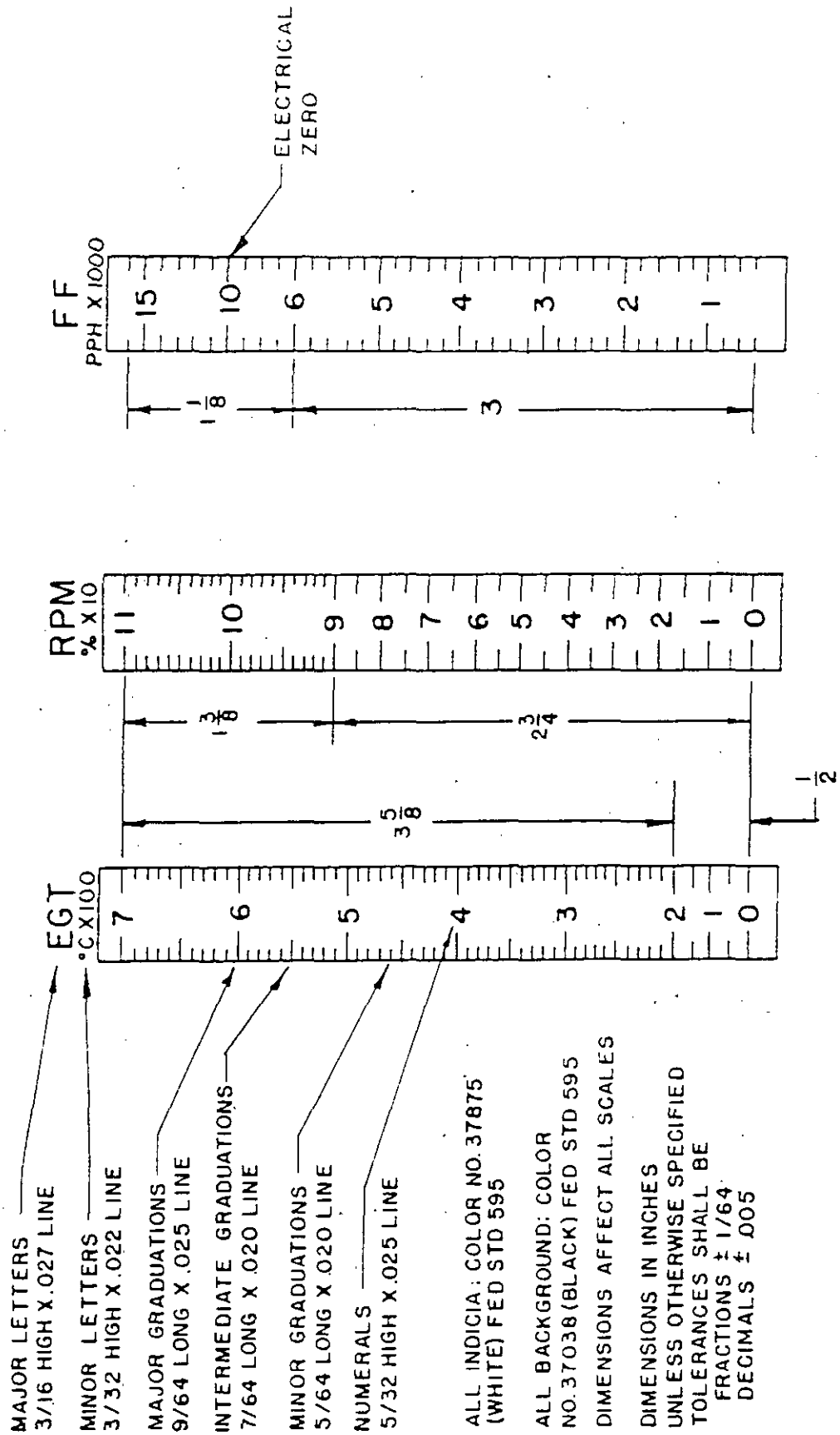


FIGURE 10 PRESENTATION AND MARKING

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d. Resolution - 0.05 percent.

3.10 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-T-31000 shall govern the manufacturer's part numbers and changes thereto.

3.11 Electronic components. Electronic components and the application thereof shall be in accordance with MIL-E-5400.

3.12 Weight.

3.12.1 Converter. The weight of the converter shall not exceed 38 pounds

3.12.2 Indicators. The weight of each indicator shall not exceed four pounds.

3.13 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practical, in accordance e with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in place of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used.

3.13.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625 except as follows:

a. Dials, small holes, and case inserts need not be anodized

b. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541

c. When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used instead of anodizing

d. Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing

e. Parts in a hermetically sealed container need not be anodized

f. When the part is plated with tin over a copper flash, the part need not be anodized.

3.13.2 Paint. Cases for the components, the background of the indicator dials, mounting bezels, gaskets, spacer rings, and nonindicating portions of the tape shall be finished in durable dull black, Color No. 37038, FED-STD-595.

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3.13.3 Anti reflection coating. All reflecting glass surfaces shall be provided with a reflection reducing coating which meets the requirements of MIL-C-675, in addition to withstanding the environmental conditions specified herein, except that the following reflectance tolerances shall apply:

Angle of Incidence	Wave Length Millimicrons	Average Percent Reflectance
0°	450 to 675	0.6
0°	425 to 700	0.5
30°	450 to 625	1.0
30°	425 to 700	0.5

Coated surfaces shall not chip, scale, peel, flake, dissolve, discolor, or otherwise be adversely affected in normal service use. The coating on the outer surface of the cover glass shall withstand abrasion incident to normal service use.

3.14 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.15 Workmanship. The converter and indicators shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine screw assembly, welding, brazing, and freedom of parts from burrs and sharp edges.

3.15.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be close as is consistent with best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the system, they shall be held or limited accordingly.

3.15.2 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt threads.

3.15.3 Riveting. Riveting operations shall be carefully performed to ensure that the rivets are tight and satisfactorily headed.

3.15.4 Cleaning. The converter and indicators shall be thoroughly cleaned of loose, spattered or excess solder, metal chips or other foreign material after assembly. Burrs and sharp edges, as well as resin flash which might crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in this specification.

4.3.1 Atmospheric. Unless otherwise specified, all tests shall be conducted at atmospheric pressure (approximately 29.92 inches Hg), at room temperature (approximately +25° C), and a maximum of 50 percent relative humidity. When tests are made with atmospheric pressures or temperatures substantially different from the above values, proper allowance shall be made from the specified condition.

4.3.2 Attitude. Unless otherwise specified, the system components shall be tested in the normal operating position.

4.3.3 Vibration. Unless otherwise specified, the indicators shall not be subjected to any form of vibration or tapping during the performance of any of the tests specified in this specification.

4.3.4 Power supply. Unless otherwise specified, the system shall be tested with 115V 400 cps a-c power supply. The voltage and frequency shall be maintained within ± one percent.

4.3.5 Converter input signals. Unless otherwise specified, the converter shall be electrically connected to input signal generators. Millivoltage inputs shown in Table IV may be used as EGT signal inputs. Synchro inputs from calibrated synchro transmitter test standards may be used for rate of flow inputs. The synchro transmitters used to provide converter input functions shall be accurate to within ±0.1° either by selection or calibration. Tachometer inputs, however, shall be furnished by the GEU-7/A tachometer generator conforming to MIL-G-26611, unless prior approval for use of an alternate method is obtained from the procuring activity.

4.3.6 Converter output signal. Unless otherwise specified, when tests of the converter are required, the converter shall be electrically connected to the applicable indicator or to a special test receiver equipped with a calibrated pointer and dial so that the pointer position versus the corresponding function reading can be readily determined. Where a special receiver is used, its error shall be known and shall not exceed 0.1 percent of full scale deflection. Converter and indicator tests are listed separately herein to provide specific test procedures for the individual components when procured separately. In the event of a system procurement, the applicable indicators shall be used as the converter output receivers and shall be subjected to the same tests as a system, the combined converter and indicator tolerances shall apply. In the event procurement is for indicators only, the input to the indicators shall be supplied by a special signal generator which shall simulate the output from the converter. The accuracy of the signal generator shall be within 0.1 percent of full scale deflection. The test equipment shall not load or affect normal operation of the unit under test. The tolerances specified are acceptable limits of performance and are not to be confused with meter accuracies.

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4.3.6.1 Cross reference data. The contractor shall devise a detailed method of testing to apply in instances where complete tests of the converter and the indicators must be performed separately. Data concerning this detailed method of testing shall be furnished as a part of the test data required in 4.4.2. The data will be used by the Government in procuring and testing spare parts of the system instead of the complete system. The contractor shall determine and furnish data which shall show converter output signal characteristics necessary to drive each type of indicator and cause it to indicate the value shown for each test point specified in Table II.

4.4 First article inspection.

4.4.1 Test samples. The test samples shall consist of two systems as described in 3.2, representative of the production equipment. The components in the system shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 First article inspection. The first article inspection shall consist of all tests described under 4.6.

4.4.3 First article inspection schedule. The first article inspection schedule shall be as listed in Table V for the converter, Table VI for the indicators, and Table VII for the system. The first article inspection shall be conducted in the sequence specified in the applicable tables unless prior approval for use of an alternate schedule is obtained from the contracting activity.

4.4.4 Fault analysis. A detailed fault analysis showing compliance with the requirements specified in 3.5.3, 3.5.8, 3.8.1, 3.8.2.5.1, and 3.9 shall be prepared by the contractor and submitted to the contracting activity.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of:

a. Individual inspections

b. Sampling plans and inspections.

4.5.1 Individual inspections. Each converter and indicator produced shall be subjected to the following tests as described under 4.6.

4.5.1.1 Converter.

a. Examination of product

b. Scale error at room temperatures.

4.5.1.2. Indicators.

a. Examination of product
b. Electrical zero

c. Scale error at room temperature

d. Short friction error
e. Short position error

f. Integral lighting

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(visual comparison)

TABLE V. Converter Test Schedule

PARAGRAPH NUMBER	TEST	TEST SAMPLE # 1	TEST SAMPLE # 2
4.6.1.1	Examination of product	x	x
4.6.1.2	Scale error at room temperature	x	x
4.6.1.3	Scale error at high temperature	x	x
4.6.1.4	High temperature exposure		x
4.6.1.5	Vibration		x
4.6.1.5.1	Tape oscillation		x
4.6.1.5.2	Tape variation		x
4.6.1.5.3	Scale error after vibration		x
4.6.1.6	Shock tests		x
4.6.1.7	Dielectric strength		x
4.6.1.8	Humidity		x
4.6.1.9	Fungus		x
4.6.1.10	Salt spray		x
4.6.1.11	Sand and dust		x
4.6.1.12	Sealing		x
4.6.1.13	External pressure		x
4.6.1.14	Scale error after testing	x	x

TABLE VI. Indicator test schedule

PARAGRAPH NUMBER	TEST	TEST SAMPLE # 1	TEST SAMPLE # 2
4.6.2.1	Examination of product	x	x
4.6.2.2	Electrical zero	x	x
4.6.2.3	Scale error at room temperature	x	x
4.6.2.4	Scale error at high temperature	x	
4.6.2.5	High temperature exposure		x
4.6.2.6	Friction error	x	x
4.6.2.7	Position error	x	x
4.6.2.8	Vibration *		x
4.6.2.8.1	Scale error after vibration *		x
4.6.2.9	Magnetic effect *		x
4.6.2.10	Thermal shock cycling *		x
4.6.2.11	Sealing *		x
4.6.2.12	Integral lighting	x	x
4.6.2.13	Salt spray *		x
4.6.2.14	Dielectric strength *		x
4.6.2.15	Decompression *		x
4.6.2.16	External pressure *		x
4.6.2.17	Scale error after testing	x	x

*These tests shall be performed using one each of the three types of indicators specified in 3.2.

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TABLE VII. System test schedule

PARAGRAPH NUMBER	TEST	TEST SAMPLE # 1	TEST SAMPLE # 2
4.6.3.1	Room temperature accuracy and functional interchangeability	x	x
4.6.3.2	System malfunction influence		x
4.6.3.3	System voltage and frequency variation	x	x
4.6.3.4	System scale error at low temperature	x	
4.6.3.5	System warm-up		x
4.6.3.6	System response	x	x
4.6.3.7	System sensitivity	x	x
4.6.3.8.1	Decreasing temperature		x
4.6.3.8.2	Increasing temperature		x
4.6.3.8.3	Thermocouple lead resistance		x
4.6.3.8.4	Grounding of thermocouple extension wires		x
4.6.3.8.5	Stray signal and beat frequency tolerance		x
4.6.3.9	System extreme low temperature exposure		x
4.6.3.10	System high altitude		x
4.6.3.11	System high altitude-low temperature		x
4.6.3.12	System radio noise interference	x	
4.6.3.13	System reliability	x	
4.6.3.14	Maintainability		x
4.6.3.15	Component compatibility and interchangeability		x

4.5.2 Sampling plans and tests.

4.5.2.1 Sampling plan A. Two systems, as described in 3.2 shall be selected at random from each 100 systems or fraction thereof produced or three samples of each component shall be selected at random from each 100 components or fraction thereof produced and subjected to the following tests as described under 4.6:

4.5.2.1.1 Converter.

a. Individual tests

b. Scale error at high temperature

c. Dielectric strength

d. Sealing.

4.5.2.1.2 Indicators.

a. Individual tests

b. Scale error at high temperature

c. Friction error

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d. Position error

e. Vibration

f. Scale error after vibration

g. Sealing

h. Dielectric strength

i. Integral lighting (sampling plan A of MIL-L-27160).

4.5.2.1.3 System.

a. System voltage and frequency variation

b. System scale error at low temperature

c. System warm-up

d. System response

e. System sensitivity

f. Thermocouple lead resistance

g. Grounding of thermocouple extensions wires

h. Stray signal and beat frequency tolerance.

4.5.2.2 Sampling plan B. Two systems, as described in 3.2, or components, as applicable, shall be selected at random from the first 10 systems or components produced and subjected to the following tests as described under 4.6:

4.5.2.2.1 Converter.

a. Sampling plan A tests

b. High temperature exposure

c. Shock tests

d. Vibration

e. Tape oscillation

f. Tape variation

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g. Scale error after vibration

h. Humidity

i. Fungus

j. Salt spray

k. Sand and dust

l. External pressure

m. Scale error after testing.

4.5.2.2.2 Indicators.

a. Sampling plan A tests

b. High temperature exposure

c. Magnetic effect

d. Thermal shock cycling

e. Salt spray

f. Integral lighting (sampling plan B of MIL-L-27160)

g. Decompression

h. External pressure

i. Scale error after testing.

4.5.2.2.3 System.

a. Room temperature accuracy and functional interchangeability

b. System malfunction influence

c. Decreasing temperature

d. Increasing temperature

e. System extreme low temperature exposure

f. System high altitude

g. System high altitude - low temperature

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h. System radio noise interference

i. System reliability

j. Component compatibility and interchangeability

4.5.2.3 Reject and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods.
4.6.1 Converter tests.

4.6.1.1 Examination of product. Each converter shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.1.2 Scale error at room temperature. The converter shall be connected to indicators having a known accuracy or a test receiver and subjected to a scale error at room temperature test by applying input signals to the converter corresponding to the test points specified in Table I. The temperature shall be maintained within $25^{o} \pm 2^{o}$ C. The tolerances shall not exceed the room temperature tolerances specified in Table I. This test shall be conducted both up scale and down scale.

4.6.1.2.1 Short scale error. When a short scale error test is specified, the test points shall be those indicated as "E" in Table I.

4.6.1.3 Scale error at high temperatures. The converter shall be placed in a test chamber and subjected to a temperature of $100^{o} \pm 2^{o}$ C for a period of four hours. While still at $100^{o} \pm 2^{o}$ C, the converter shall be subjected to a short scale error test (see 4.6.1.2.1). High temperature scale errors shall not exceed the tolerances specified in Table I.

4.6.1.4 High temperature exposure. The converter shall be placed in a test chamber and subjected to a temperature $100^{o} \pm 2^{o}$ C for a period of 24 hours. Normal instrumentation power shall be applied continuously the last two hours of this period. While the converter is still at this temperature, the test receiver or indicators shall be operated throughout the parameter ranges without exhibiting erratic characteristics. The converter shall then be subjected to a short scale error test (see 4.6.1.2.1). High temperature scale errors shall not exceed the tolerances specified in Table I. The converter shall then be returned to room temperature for a minimum period of four hours and subjected to a short scale error test. The scale errors shall not exceed the room temperature tolerances specified in Table I.

4.6.1.5 Vibration. The converter shall be subjected to the vibration test specified in MIL-STD-810, Method 514.2, procedure I, parts 1 and 2, curve B. Electronic assemblies shall be tested per part 2. The indicators shall be connected to the converter during this test and the system shall be energized. The indicators shall be observed periodically throughout the test for tape oscillation and tape variation as specified below.

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4.6.1.5.1 Tape oscillation. The maximum amplitude of the indicator tape oscillation throughout the frequency range while the converter is vibrated shall be noted and shall not exceed the tolerance shown in Table I.

4.6.1.5.2 Tape variation. The maximum variation of the indicator tape, during ECK-25 converter vibration, from its original position before vibration shall be noted throughout the frequency range. The maximum variation of the indicator tapes shall not exceed the tolerance shown in Table I.

4.6.1.5.3 Scale error after vibration. Following the vibration period, the converter shall be tested in accordance with 4.6.1.2.1. Allowable tolerances specified in Table I shall apply.

4.6.1.6 Shock tests. The converter shall be subjected to the shock test in accordance with MIL-STD-810, Method 516.2, procedure I. Following the shock test, a room temperature short scale error test shall be conducted. The errors shall not exceed the tolerances specified in Table I.

4.6.1.7 Dielectric strength. The converter shall be tested for dielectric strength. A voltage of 1,250V ac rms at commercial frequency shall be applied for one minute to all parts of nonhermetically sealed converters that carry 65V to 115V. A voltage of 500V ac rms at commercial frequency shall be applied for one minute to all parts of nonhermetically sealed converters that carry 65V or less. Transistors and similar sensitive parts may be disconnected from the circuit prior to testing. A voltage of 200V ac rms at commercial frequency shall be applied to hermetically sealed converters for one minute. In the event hermetically sealed converters are unsealed to disconnect transistors or sensitive parts, they need not be resealed but shall be tested as nonhermetically sealed converters.

4.6.1.8 Humidity. The converter, except when completely hermetically sealed, shall be subjected to the humidity test in accordance with procedure I of MIL-STD-810, Method 507.1, except at 95 percent humidity.

4.6.1.9 Fungus. The converter, except when completely hermetically sealed, shall be subjected to the fungus test in accordance with procedure I of MIL-STD-810, Method 508.1.

4.6.1.10 Salt fog. The converter shall be subjected to a salt fog test in accordance with procedure I of MIL-STD-810, Method 509.1, except as specified herein. At the end of the 50-hour test period, the converter shall be visually inspected and a record made of any damage resulting from this test. Salt deposits resulting from the exposure conditions may be removed by rinsing with tap water, and the converter may then be placed in a test chamber with the temperature maintained at 71° C for a period not exceeding six hours. At the end of this period, instrumentation power shall be applied to the converter and the test receiver or indicators shall be operated throughout the parameter ranges without exhibiting erratic characteristics. After an additional 48-hour period, the converter shall be subjected to the short scale error test specified in 4.6.1.2.1. The scale errors shall not exceed the room temperature tolerances specified in Table I.

4.6.1.11 Dust (fine sand). The converter shall be subjected to a dust (fine sand) test in accordance with procedure I of MIL-STD-810, Method 510.1. Upon completion of the test, the converter shall be subjected to the short scale error test specified in 4.6.1.2.1. The scale errors shall not exceed the room temperature tolerances specified in Table I.

4.6.1.12 Sealing. The converter, if hermetically sealed, shall be tested by means of a mass spectrometer type of helium leak detector. Where a nitrogen-helium mixture is utilized as the filling medium, the detected leak rate shall not exceed 1.0 micron cubic foot per hour at a pressure differential of approximately one atmosphere. Where 100 percent helium is utilized as a filling medium, the leak rate shall not exceed 10.0 micron cubic foot per hour.

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4.6.1.13 External pressure. An absolute pressure of 56 inches Hg (approximately 26 inches Hg above atmospheric) shall be applied for a period of 15 minutes with rated power on the converter channels. Normal operation shall not be affected and there shall be no failures of hermetic seals nor shall there be any case distortion

4.6.1.14 Scale error after testing. At the completion of all converter tests, the converter shall be subjected to the test specified in 4.6.1.2. Allowable tolerances for room temperature as specified in Table I shall apply.

4.6.2 Indicator tests.

4.6.2.1 Examination of product. Each EHK-19, ERK-9, and EFK-18 indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2.2 Electrical zero. The electrical zero for the rate of flow indicators shall be checked in accordance with procedures outlined in MIL-T-27179. Upon application of power, the top of the white tape column shall align itself with the longitudinal center of the electrical zero reference mark ($\pm$ 0.005 inch) shown on figure 10. The procedure will be repeated until all four indicator tapes of each indicator have been checked for electrical zero.

4.6.2.3 Scale error at room temperature. The input signal values applicable to the test points specified in Table II shall be applied to the indicators by means of a master converter having a known accuracy or a special signal generator. The temperature shall be maintained within 25 ± 2° C. The scale errors of the indicators shall be noted at the specified test points and the tolerances shall not exceed those specified in Table II. This test shall be conducted both up scale and down scale. During this test, the indicators shall be checked for stop location and off scale indications as specified in 3.9.9 and 3.9.10 respectively.

4.6.2.3.1 Short scale error test. When a short scale error test is specified, the test points shall be those indicated as "E" in Table II.

4.6.2.4 Scale error at high temperature. The EHK-19, ERK-9, and EFK-18 indicators shall be placed in a test chamber and subjected to a temperature of $+71^{o} \pm 2^{o}$ C for a period of four hours. At the end of this period and while still at this temperature, the indicators shall be subjected to the short scale error test (see 4.6.2.3.1). The scale errors shall not exceed the tolerances specified in Table II for scale error at high temperature.

4.6.2.5 High temperature exposure. The indicators shall be placed in a chamber and subjected to a temperature of $71^{o} \pm 2^{o}$ C for a period of 24 hours. Normal instrumentation power shall be applied continuously the last two hours of this period. While still at this temperature, the indicators shall be operated throughout their ranges and shall exhibit no sticking or erratic characteristics. The indicators shall then be subjected to and shall meet the tolerance requirements of the scale error at high temperature test. The indicators shall then be returned to room temperature for a minimum period of two hours and subjected to a short scale error test. The scale errors shall not exceed the tolerances specified in Table II for room temperature. The indicators shall then be subjected to a response test. The response time shall not exceed six seconds for full scale deflection.

4.6.2.6 Friction error. The friction error test may be conducted in conjunction with the scale error at room temperature test at the test points specified in Table II. The difference in indication shall be noted after a slow change in increasing indication before and after the indicator is lightly tapped. The procedure shall be repeated for decreasing
indications. The maximum observed difference in indication shall not exceed the tolerance allowed for friction in Table II. The movement of the indicator tapes shall be smooth when the millivoltage is varied uniformly.

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4.6.2.7.1 Short position error. The short position error test shall be conducted by rotating the indicator around the major axis from the normal position 0° to 180°. The reading shall not change from the set values by more than the allowable tolerances specified in Table II.

4.6.2.8 Vibration. The indicators shall be subjected to a vibration test per MIL-STD-810, Method 514.2, procedure I, part 2 only, curve B. During cycling, each type of indicator shall be so energized that its normal tape position will indicate 450° C EGT, 100 percent rpm, or 5,000 pph rate of flow, as applicable. At no time shall tape oscillation or position deviate from the above respective tape positions by more than $\pm 1^{o}$ C EGT, $\pm$ 0.25 percent rpm, or $\pm$15 pph rate of flow, as applicable. No condition of applied vibration shall cause any apparent change in the "OFF" conditions. Any condition that would result in false indication of the power-off warning positions will be unacceptable. No malfunction of the indicators shall result from this test.

4.6.2.8.1 Scale error after vibration. Following the vibration test, the indicators shall be tested in accordance with 4.6.2.3.1. Friction error shall also be checked during this test. Allowable tolerances specified in Table II shall apply.

4.6.2.9 Magnetic effect. The indicators, operating on rated voltage, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicators $5 \pm 1/2$ inch from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicators shall be held in positions 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, each indicator shall be rotated about its own horizontal axis until it is in its normal viewing position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 5°

4.6.2.10 Thermal shock cycling. The indicators shall be subjected to a total of eight cycles of immersion in tap water at a temperature of $85^{o} \pm 2^{o}$ C and then in tape water at a temperature of $5^{o} \pm 2^{o}$ C. The length of time for each bather immersion shall be 30 minutes with not more then three seconds between immersions. No evidence of moisture penetration or damage to the case shall result from this test. Following this test, the indicators shall be subjected to and shall meet the test specified in 4.6.2.11. Fogging of the cover glass during any part of this test shall be cause for rejection.

4.6.2.11 Sealing. The hermetically sealed indicators shall be tested for leaks by means of a mass spectrometer type of helium-leak detector. Where a nitrogen-helium mixture is utilized as the filling medium, the detected leak rate shall not exceed 1.0 micron cubic foot per hour at a pressure differential of approximately one atmosphere. Where 100 percent helium is utilized as the filling medium, the leak rate shall not exceed 10.0 micron cubic foot per hour.

4.6.2.12 Integral lighting. Integral lighting tests shall be conducted in accordance with MIL-L-27160.

4.6.12.1 Visual comparison. Indicator lighting shall be checked by comparison with a prime standard. The prime standard and the instrument under test shall be placed side by side at eye level. The brightness of the two units shall be compared with 2.85 ± 0.1V applied to both the standard and the unit under test. General brightness lighting level comparison shall then be made between the indicator being tested and the prime standard, with the prime standard voltage at 2.7V and then at 3.0V. If the general brightness level falls within the brightness limits of the prime standard when energized at the specific voltage levels, the indicator shall be considered acceptable. If any area of the instrument under test appears visually to differ from the same area of the prime standard, it shall then be subjected to the brightness tests specified in MIL-L-27160, and sufficient data recorded to substantiate conformance with lighting requirements.

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4.6.2.12.2 Color comparison. A color comparison shall be made by applying 4.5V to the prime standard and to the indicator under test. The two indicators shall be compared and shall visually appear identical in color. If the instrument under test appears to be a different color than the prime standard, it shall be subjected to the color tests specified in MIL-L-27160.

4.6.2.13 Salt fog. The indicator shall be subjected to a salt fog test in accordance with MIL-STD-810, Method 509.1, procedure I. The short scale error test shall be conducted immediately before and after this test. The scale error tolerances shall not exceed those specified for room temperature. Subsequent indicator operation shall not be affected by this test, and excessive corrosion of metal parts, including electrical receptacles, shall be cause for rejection.

4.6.2.14 Dielectric strength. The indicators shall be tested for dielectric strength. A voltage of 200V ac rms at commercial frequency shall be applied to hermetically sealed indicators for one minute. In the event that the indicators are unsealed to disconnect transistors or sensitive parts they need not be resealed but shall be tested as nonhermetically sealed indicators per 4.6.1.7.

4.6.2.15 Decompression. The indicators shall be placed in a cabinet in which the pressure is maintained at the same pressure as inside the hermetically sealed case. The pressure within the cabinet shall be released in such a manner that a pressure differential of 16 inches Hg will be instantaneously applied between the interior and the exterior of the case. Operation of the indicators shall not be affected by the sudden decrease in pressure.

4.6.2.16 External pressure. The indicators shall be subjected to an absolute pressure of 56 inches Hg (approximately 26 inches Hg above atmospheric) for a period of 15 minutes with rated power on the indicators. Normal operation shall not be affected by this tests. There shall be no failure of the cover glass or hermetic seals, nor shall there be any case distortion.

4.6.2.17 Scale error after testing. At the completion of all indicator tests, the indicators shall be tested in accordance with 4.6.2.3. Allowable tolerances for room temperature as specified in Table II shall apply.

4.6.3 Systems tests.

4.6.3.1 Room temperature accuracy and functional interchangeability. The engine instrumentation system shall be subjected to a room temperature accuracy and functional interchangeability tests. Input signals applicable to the test points specified in Table I shall be applied to the converter. For the system, the tolerances shall not exceed the combined room temperature tolerances specified in Table I for the converter and Table II for the indicator. The temperature shall be maintained within 25 ± 2^{O} C. This test shall be conducted both up scale and down scale. All indicators shall then be connected to the second converter and the above room temperature accuracy test repeated. In no instance shall the system tolerance exceed the combined tolerances specified in Table I and II.

4.6.3.2 System malfunction influence. Starting with the complete system connected and functioning, the system shall demonstrate its ability to keep the remainder of the system operating properly when other portions malfunction. Removal of the main 115V 400-cycle input power from one, two, or three of the engine instrument groups shown on Figure 1 shall have no effect upon the proper operation of any other instrument channel not included in those groups. Furthermore, no power failure or malfunction within any one channel circuitry shall cause a malfunction in any other channel. Each indicator power-off warning shall be observed and shall give proper indication of power failures during this test.

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4.6.3.3. System voltage and frequency variation. The system shall be subjected to a short scale error at room temperature test at test points indicated as "E" in Table I and the input power varied to provide any combination of the following voltages and frequencies. The scale errors of the system at the varied voltages and frequencies shall not exceed the combined scale errors specified in Tables I and II.

a. 104V to 126V

b. 380 to 420 cps.

4.6.3.4 System scale error at low temperature. The system shall be placed in a test chamber and subject to -54° ± 2° C for a period of four hours. At the end of this period and while still at this temperature, input signals applicable to the test points indicated as "E" in Table I shall be applied to the converter. The scale errors shown by the indicators shall not exceed the combined tolerances specified in Tables I and II for scale error at low temperature.

4.6.3.5 System warm-up. Applicable input signals shall be applied to the converter along with 115V, 400 cycle excitation power. The input devices shall be set to provide 100° c EGT, 10 percent rpm, and 1,000 pounds per hour readings on the EGT, tachometer, and rate of flow indicators, respectively, and the excitation power then turned off. The respective input devices shall then be set to a value that will produce full scale indication on the indicators. After the excitation power has been off for at least one hour, it shall be turned on and the time required for the indicators to reach full scale deflection and to indicate within the accuracies specified in Tables I and II for room temperature shall not exceed one minute.

4.6.3.6 System response. Input signals shall be applied to all parameters of the converter and all functions set to indicate half scale on the output indicators. The EGT and rate of flow input signals shall then be instantaneously increased to a value that will produce full scale indication. The time required to reach full scale deflection shall not exceed three seconds. There shall be no noticeable lag between the tachometer input and output when the tachometer input is gradually increased from half scale to 110 percent in seven seconds.

4.6.3.7 System sensitivity. Input signals shall be provided to the converter which shall produce 400° C EGT, 100 percent rpm, and 4,000 pph rate of flow. The input signals shall then be increased by 2° C EGT, 0.25 percent rpm, and ten pph respectively. The resultant output readings shall be changed as follows: 2° ± 0.5° C EGT, 0.25 ± 0.12 percent rpm, and 10 ± 5 pph, respectively.

4.6.3.8 System cold junction compensator lag and lead resistance - EGT section only.

4.6.3.8.1 Decreasing temperature. The EGT portion of the converter shall be subjected to a relatively constant room temperature for a period of not less than three hours. The converter shall then be connected to a chromel-alumel thermocouple lead with the thermocouple maintained at 0° C and a constant millivoltage applied in series with the thermocouple lead to provide an initial indication of 400° C. The converter shall then be placed in a test chamber, the temperature of the chamber decreased to -54° ± 2° C at a rate of approximately 5° C per minute, and the reduced temperature held for a period of one hour. The reading of the indicators, when the temperature is being reduced shall at no time vary from the 400° C reading by more than 5° C. After one hour at -54° ± 2° C, the indicator reading shall be 400° ± 10° C,

4.6.3.8.2 Increasing temperature. This test shall be conducted in accordance with 4.6.3.8.1 except the temperature shall be raised from room temperature to 100° ± 2° C at a rate of approximately 5° C per minute. The tolerance specified in 4.6.3.8.1 shall apply.

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4.6.3.8.3 Thermocouple lead resistance. Power shall be applied to the indicators and the indicators subjected to a lead resistance change from 5 to 200 ohms at the 300°, 400°, 500°, and 700° C, test points. The change in indication shall not exceed 2° C.

4.6.3.8.4 Grounding of thermocouple extension wires. The chromel or plus wire and the alumel or minus wire shall be alternately connected to the grounded or low-potential side of the 155V, 400-cycle power supply. Prior to the grounding of a wire, the scale error shall be determined at the 300° and 400° C test points and the response time determined in the 200° to 500° C scale range on increasing temperature. The scale error and response tests shall be conducted with each lead alternately grounded. The scale errors with either lead grounded shall not differ from the scale error without a lead grounded by more than 2° C. The response time with the leads alternately grounded shall not differ from the response time without a lead grounded by more than one second.

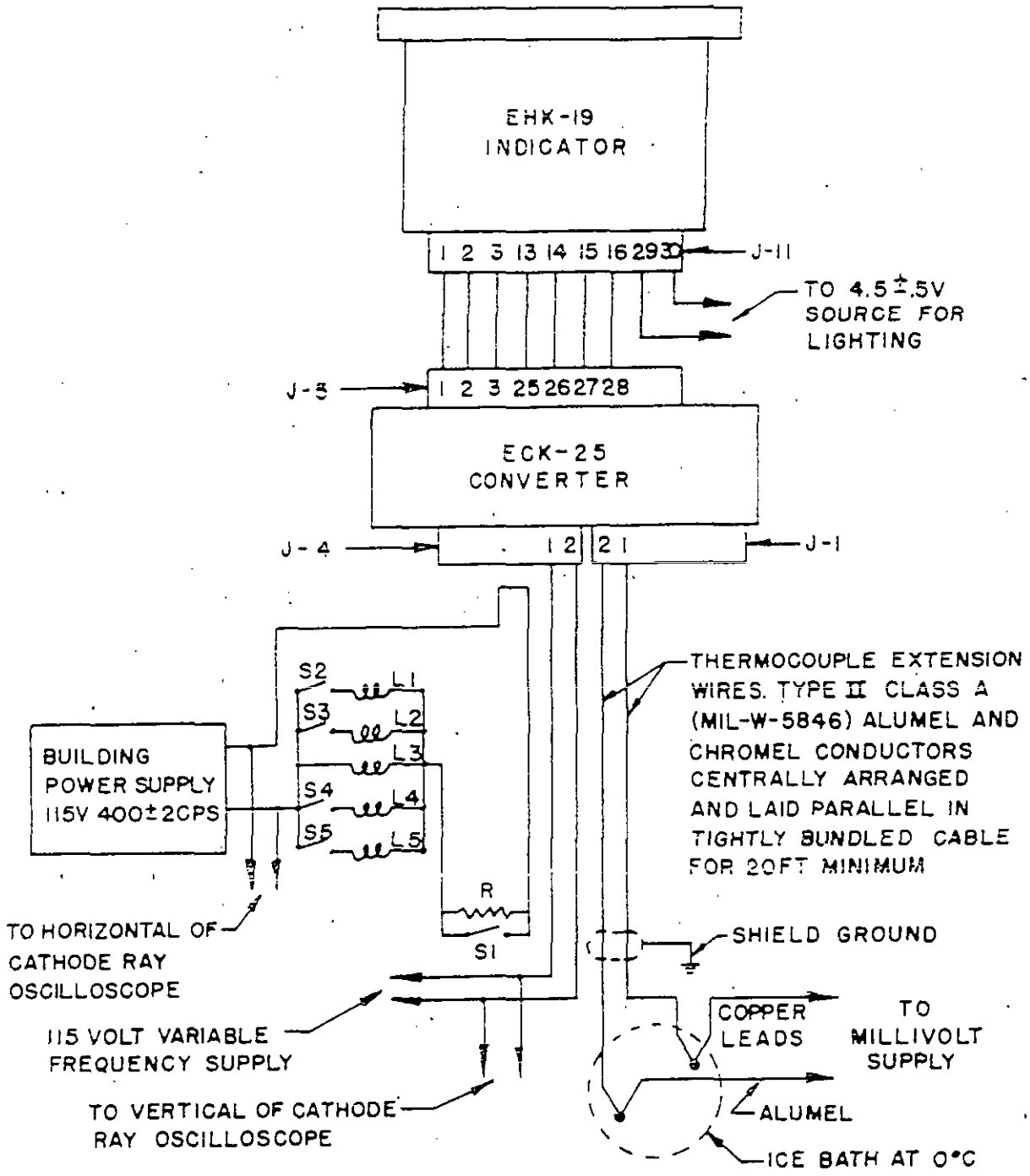
4.6.3.8.5 Stray signal and beat frequency tolerance. The ECK-25 converter and EHK-19 indicator shall be connected as shown in Figure 11 and all switches opened. With the building power turned off, 115V 400-cycle power and a simulated thermocouple signal to produce an indication of 450° C shall be applied to the indicator. After two minutes at the 450° C indication, the cathode ray oscilloscope and the building power supply shall be turned on and allowed to warm up for an additional two minutes. Beginning with a frequency of 380 cps, the frequency of the variable-frequency supply shall be slowly increased a maximum of one cps each five seconds until 420 cps is reached. The temperature indication shall be closely observed throughout the test. Any change in indicator tape position, oscillation of the tape, or sluggishness shall be recorded and the frequency value noted. This procedure shall be repeated with only S-1 closed; S-1 and S-2 closed; S-1, S-2, and S-3 closed; S-1, S-2, S-3, and S-4 closed; and then with all switches closed. During this test, each of the thermocouple leads shall be alternately grounded with each group of switches closed. Tape oscillation shall be cause for rejection. The maximum change in tape position throughout the test shall not exceed 2° C. If sluggishness is observed during any of the above conditions, the response time test shall be conducted with the worst condition maintained. The response time shall not exceed 3.0 seconds.

4.6.3.9 System extreme low temperature exposure. The system shall be placed in a test chamber and subjected to a temperature of -65° ± 2° C for a period of 48 hours. While still at this temperature, the inputs to the respective converter channels shall be operated throughout their applicable ranges without exhibiting erratic characteristics in the indicators. The temperature shall be raised to -54° ± 2° C and maintained for 24 hours. While at this temperature, the system shall be subjected to the warm-up and response time tests specified herein. The warm-up time required to produce specified accuracies shall not exceed two minutes and the response time shall not exceed ten seconds. The system shall then be subjected to the system scale error at low temperature test specified in 4.6.3.4 and the scale errors shown by the indicators shall not exceed the combined tolerances specified in Tables I and II for low temperature. The system shall be returned to room temperature for a minimum period of four hours and subjected to a short scale error at room temperature test at test points indicated as "E" in Table I. The scale errors shall not exceed the combined tolerances specified in Tables I and II for room temperature.

4.6.3.10 System high altitude. The system shall be subjected to an absolute pressure of 2.12. ±0.1 inch Hg, equivalent to an altitude of approximately 60,000 feet, for a period of one hour. While still at this pressure, the system shall be subjected to a short scale error at room temperature test at test points indicated as "E" in Table I. The scale errors shall not exceed the combined tolerances specified in Tables I and II for room temperature. There shall be no damage to the system as a result of this test.

4.6.3.11 System high altitude - low temperature. The procedure specified in 4.6.3.10 shall be repeated except that the temperature shall be lowered in 4.6.3.10 shall be repeated except that the temperature shall be lowered to -54° ± 2° C. The scale error shall not exceed the combined tolerances specified in Tables I and II for low temperature. There shall be no damage to the system as a result of this test.

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R = 100,000 ± 5,000 OHMS
L1, L2, L4, L5 = 300 WATT LAMPS OR EQUIVALENT IMPEDANCES
L3 = 60 WATT LAMP OR EQUIVALENT IMPEDANCE

FIGURE 11. Stray signal and beat frequency test

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TABLE VIII.

MAXIMUM ALLOWABLE FAILURES	TOTAL TEST TIME (HOURS)
0	1,500
1	1,780
2	2,040
3	2,310
4	2,580

4.6.3.12 System radio noise interference. The system shall be subjected to a transmitted and susceptibility noise suppression test in accordance with MIL-I-6181.

4.6.3.13 System reliability. The system shall be subjected to a reliability test in accordance with the following. The test time and maximum allowable failures shall be in accordance with Table VIII. The system shall be tested for at least 1,500 hours of actual electrical operation. During this time, input signals shall be applied to the converter for 50,000 cycles of operation at a rate of 4 ± 1 cycle per minute as follows: EGT parameter from 0° to 550° and return; percent rpm parameter from 10 to 90 percent rpm and return; and rate of flow parameter from 500 to 12,800 pph and return. The test shall begin when rated electrical power is applied. After completion of four hours of operation with rated power on, which shall include 135 cycles, the power shall be turned off and shall remain off for a period of 1/2 hour. The power-on power-off procedures shall be repeated until the specified 1,500 hours and 50,000 cycles of electrical operation have been achieved. In case of malfunction, the malfunction shall be recorded and the test interrupted while repairs are made. The test shall then be continued from the point of interruption. Upon completion

of each 250 hours of cycling operation, the system shall be checked for scale error. The scale error readings shall not differ by more than one percent of full scale from those obtained at room temperature previous to this test. No recalibration shall be required to pass the error test. This test may be interrupted for normal weekend shutdowns. Satisfactory completion of this test will demonstrate an MTBF of at least 652 hours at a 90 percent confidence level.

4.6.3.14 Maintainability verification. Maintainability verification shall be in accordance with MIL-STD-471, appendix A, paragraph entitled "Maintenance Task Selection."

4.6.3.15 Component compatibility and interchangeability. Demonstration of component compatibility and functional interchangeability as specified in section 3 herein shall be conducted in accordance with the procedures specified in the applicable contract document for the engine instrumentation system specified herein.

5. PACKAGING

5.1 Preservation. Preservation shall be level "A" or "C" as specified (see 6.2).

5.1.1 Preservation-packaging. Unless otherwise specified by the contracting activity, item shall be packaged in Quantity Unit Pack (QUP) of one each. Each item will be provided a preservation method IC-1 in accordance with MIL-P-116.

5.1.2 Cleaning. Indicator shall be cleaned in accordance with process C-1 of MIL-P-116.

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5.1.3 Drying. Immediately after cleaning, the item shall be dried following any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved in accordance with MIL-P-116 and MIL-STD-2073-1 to provide a method IC-1.

5.1.5 Level C. Item shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073-1.

5.2 Packing. Packing shall be level "A," "B," or "C" as specified (see 6.2).

5.2.1 Container-Cushioning. Unit container requirements shall conform to MIL-B-121, barrier material, unless otherwise specified by the contracting activity. Unit container shall be specified by the contracting activity. Unit container shall be large enough to allow for the application of cushioning/wrap material, of sufficient density and thickness, to protect the unit container from punctures. Intermediate/shipping containers shall be large enough to allow for the application of sufficient cushioning material, between the unit container and intermediate/shipping container. Cushioning material shall be of sufficient density and thickness to ensure shock transmission does not exceed peak values in "G's" established for the item.

5.2.2 Level A. Fiberboard containers do not meet level "A" container criteria.

5.2.3 Level B. Item will be preserved as specified in 5.1 and shall be packed in exterior containers conforming to PPP-B-1672, weather-resistant unless otherwise specified by the contracting activity. Exterior container shall be uniform shape, size, and minimum tare and cube, consistent with the protection required. Closure shall be in accordance with appropriate PPP-B-1672 procedure and contracting activity.

5.2.4 Level C. Item will be packed in such a manner that will afford adequate protection against physical damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to MIL-STD-2073-1.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked in accordance with special markings required by the contracting activity and MIL-STD-129.

5.4 Inspection and test. Test of methods of preservation shall be accomplished in accordance with section 4 of MIL-P-116 to ensure compliance with section 5 of this specification. Packaging tests shall be conducted in accordance with rough handling as specified in FED-STD-101.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The A/A24J-11 engine instrumentation system covered by this specification is intended for use on four-engine turbofan or turbojet aircraft. This system utilizes exhaust gas temperature, tachometer, and rate of flow signals to produce indications of these parameters for the pilot's and flight engineer's usage.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Number of first article samples and point of testing.

d. Levels of preservation, packaging, and packing required.

6.3 First article. When the first article inspection is required, the contracting officer should provide specific guidance to offerors. The contracting officer shall specify whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1); and the number of items to be tested as specified in 4.3.

6.4 Definitions.

6.4.1 Converter channel. A converter channel is a single, self-sufficient, electrically isolated unit (entirely contained within the ECK-25 converter assembly) that receives an input signal and provides a processed signal through
two outputs. One output drives the pilot's indicator channel and the other output drives the flight engineer's within the indicator case.

6.4.2 Hermetic sealing. Hermetic sealing is the process by which an item is totally enclosed by a suitable metal structure or case and sealed airtight by fusion of metallic or ceramic materials. This includes the fusion of metal by welding, brazing, or soldering, fusion of ceramic materials under heat or pressure; and fusion of ceramic materials into a metallic support. Elastomeric or resinous materials, or combinations of these materials, may be used as a cover glass seal provided the specified leak rate requirements are met under all test and environmental conditions specified herein. Elastomeric or resinous materials may be used in other areas provided prior approval is obtained from the procuring activity. A hermetic seal is not intended to include seals accomplished by gaskets. _

6.4.3 Indicator channel. An indicator channel is one complete vertical scale display mechanism of a quadruple vertical scale tape indicator. Each indicator channel is electrically and mechanically isolated from the other three indicator channels contained within the indicator case.

6.5 Subject term (key words) listing. Recommended key words are:

Engine
Exhaust gas temperature
Four engines
Tachometer
Turbofan or turbojet

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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	MIL-E-27669C(USAF)	94/09/15

3. DOCUMENT TITLE
Engine Instrumentation, System, Aircraft A/A24S-11

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

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